Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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***Abstract*—Wheeled bipedal robots face a fundamental trade-off: stiffness for sub-millimeter precision shrinks push-recovery capture, while integral action that cancels equilibrium bias winds up during disturbances. We present the Anchored Conditionally-Gated Controller (ACC), a two-channel torque assembly whose core mechanism is a proximity-gated anchor: a position integral confined to a 15 cm neighborhood, enabled by an asymmetric envelope follower ($\tau_{\text{attack}}\approx$23 ms, $\tau_{\text{release}}\approx$1.5 s) that distinguishes quiet stance from post-push ringdown. ACC delivers omnidirectional push recovery (F_{min}=83 N, F_{med}=117 N, 8-direction sweep, N=10) together with sub-millimeter quiet-stance precision. We report accuracy (RMS from commanded home) and repeatability (RMS about the trial mean) separately, and resolve both on the sagittal axis—the actuated inverted-pendulum direction—rather than pooling them into a planar norm diluted by an axis this class of platform cannot drive. Under idealized conditions (clean sensors, 0 ms delay, stiff contact, MuJoCo), ACC holds 0.294±0.015 mm sagittal repeatability (N=10), a 59× improvement over a proportional-only baseline measured under one unified protocol. Accuracy is a full order of magnitude worse on the same axis: the robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm, and this gap is treated as the design's principal open weakness. A factorial ablation (N=10 per row) separates the two: the gated damping terms produce the steadiness (removing the boost costs a factor of 16 in sagittal repeatability), while the anchor integral produces 4.2 mm of the accuracy at a 3× cost in steadiness; the proximity gate and asymmetric envelope are inert during quiet stance and decisive only under disturbance. Precision degrades by a factor of 5.5 at 10 ms actuation delay, and to a hardware-projected 1 – 3 mm once encoder quantization, backlash, stiction, and IMU noise are accounted for; F_{max} bifurcates from 96 N to 47 N at any non-zero delay. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate, N=20, all at ACC's 100 Hz control rate); a preliminary model-free PPO run (5.4M steps, single seed, no domain randomization) falls in 85 % of multi-height transitions—reported as a difficulty probe on unaided policy search, not as a tuned reinforcement-learning baseline; removing the PID servo path (4-state Direct Torque LQR) improves survival from 0.32 s to 0.60 s but does not prevent the fall. Flight recovery (100 cm drop, 50 cm ledge) and per-leg terrain adaptation (10 – 20 cm one-wheel curb) are demonstrated at N=10 per condition and ablated: the per-leg height split is necessary at every curb height tested (0/10 without it), whereas the airborne wheel-torque override proves to be a variance-bounding refinement rather than an enabling mechanism at these scales, a negative result. Core claims rest on $N\geq$5.**

***Index Terms*—Wheeled biped robot, conditionally-gated control, anchored standing, proximity-gated integral, push recovery, flight recovery, terrain adaptation.**

I. Introduction

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance [1], WBC posture [2], MPC terrain [3], and RL locomotion [4].

The difficulty is fundamental: lowering k_p widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of k_p (measured at N=10: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally, 49.1 mm planar RMS from the commanded pose)—the oscillation lies on the one axis the wheels can drive, which is both why it persists and why it is the axis every mechanism in this paper is aimed at—yet a global integral winds up during pushes.

Preliminary PPO [5] training (5.4M steps, single seed, in our `BalanceEnv`) showed limited height generalization: the policy stood at a single height (Surv ≈3.20 s, vs. 0.16 – 0.60 s for the classical baselines) but failed to generalize across 0.40 – 0.70 m base-z (§III-A)—height RMSE >50 mm, 85 % falls on squat transitions—with the failures concentrated on precisely the coupled pitch–height dynamics that a posture map and gate schedule encode explicitly. This run is retained throughout as a *difficulty probe*, not as a reinforcement-learning baseline. Its purpose is to characterize how hard unaided model-free search is on the coupled pitch–height state space of a 10-DOF wheeled biped, and thereby to make explicit what a structured prior supplies in its place; it is single-seed, without domain randomization or a hyperparameter search, and no claim about the attainable performance of RL on this platform is drawn from it.

We present the Anchored Conditionally-Gated Controller (ACC), addressing both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ($\tau_{\text{attack}}\approx$23 ms, $\tau_{\text{release}}\approx$1.5 s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294±0.015 mm repeatability on the sagittal axis—the actuated inverted-pendulum direction, and the one on which the limit cycle above lives—under idealized conditions (clean sensors, 0 ms delay, N=10), at push performance equal to or better than the proportional-only baseline that carries no integral action at all. Absolute accuracy on that axis is a separate and weaker result, treated in Sec. V-A: the robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm.

A factorial ablation measured under a single protocol (Table III) locates the mechanism more precisely than the gate structure suggests, and reveals a decoupling of mechanisms that contrasts with the architecture's design rationale on two of its three counts. First, the quiet-stance *steadiness* comes from the gated *damping* terms, not from the anchor integral: disabling the integral outright while leaving every gate intact leaves sagittal repeatability sub-millimeter (it in fact improves, 0.294 mm→0.089 mm), whereas reverting the two velocity-damping coefficients reinstates the full limit cycle (25.2 mm). What the integral buys is not steadiness but *accuracy*—it removes 4.2 mm of sagittal parking offset—and it buys that at a 3× cost in steadiness. The two properties are produced by different mechanisms and trade against each other, which is why they are reported as separate columns rather than as a single planar number. Second, the proximity gate and the asymmetric envelope are provably inert during quiet stance—ablating either reproduces ACC bit-for-bit—because the robot never leaves the gate's inner band nor the envelope's release branch while standing still; their contribution is windup prevention and ringdown under disturbance, and only a displacement experiment can measure it. Third, the pitch safety gate is load-bearing even with no disturbance applied: removing it causes divergence from the initial-condition noise alone in 10/10 trials. The gates are thus enabling conditions that make an otherwise inadmissible damping and integral configuration safe, rather than the proximate source of the precision. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate), confirming that platform complexity, not servo lag alone, dominates the failure. The preliminary model-free PPO reference (5.4M steps, single seed) falls in 85 % of multi-height transitions; removing the PID servo path (Direct Torque LQR) buys only 0.28 s of survival with persistent 100 % falls.

The paper is positioned within *structured classical priors* rather than within learned control. The object of study is a hand-designed, fully interpretable torque assembly whose gating encodes the plant's measured failure modes explicitly, and whose intended downstream role is to serve as the nominal base action of a residual-learning stack—the setting in which a policy corrects a structured prior instead of searching the space unaided [6], [7]. The classical controllers are therefore the comparison of record; the model-free run of §V-D is evidence about the difficulty of the unaided problem, and is not offered as a bound on either method.

The main contributions are:

1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
2) A **two-channel conditionally-gated torque assembly** with flight recovery and per-leg terrain adaptation as independent extensions, each ablated at N=10—including the negative result that the flight override is not load-bearing at the reported drop scales.
3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) with six classical baselines (four 4-state-derived and two 6-state coupled) and a preliminary model-free PPO reference (5.4M steps, single seed) included as a difficulty probe.

II. Related Work

A. Wheeled Bipedal Platforms and Classical Control

Ascento [1] demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops [2]. Handle [8] pioneered wheeled-leg coordination; wheeled quadrupeds combined driving with walking through whole-body motion control and planning [9]. Ollie [10] adapts whole-body balance control on a two-wheeled biped, and wheeled quadrupeds have since been driven by whole-body MPC with online gait sequencing [11]. DIABLO [12] pairs an LQR balance controller with separate yaw, split-angle, height, and roll loops; SKATER [3] uses a hierarchical MPC whole-body controller with centrifugal-force compensation and terrain adaptation. Whleaper [13] is closest to our morphology (10 DOF, three hip DOFs per leg); it applies LQR to balanced sliding and a separately trained PPO policy to walking, *switching* between the two by locomotion mode rather than composing them on a shared control output. Composition of a classical prior with a learned residual on one output is the residual-RL construction [6], [7], applied recently to wheeled-biped balance [14]; ACC is designed to supply the classical prior in that construction rather than to replace the learned part. Coupled pitch–roll LQR is standard on rigid two-wheeled platforms, where torque acts directly on the wheels and there are neither articulated legs nor leg-height dynamics—a structurally simpler problem with fewer unstable modes. Our 6-state LQR reproduces that coupled structure on a legged morphology and fails; the 4-state Direct Torque LQR—removing the servo path and re-optimizing roll gain ($k_{p,\mathrm{roll}}$=55.0)—also fails (100 % fall rate, 0.60 s). Because the direct-torque variant carries no servo lag at all and still falls (§V-D), actuation lag alone cannot account for the outcome; the articulated-leg morphology and its height dynamics are the dominant remaining factor. Both LQR variants use gains derived from a simplified 4-state TWIP model, not a full 10-DOF linearization of the articulated-leg plant (see Limitation 7); these results are therefore empirical—demonstrating that the specific LQR implementations tested fail—not a proof that LQR is structurally insufficient. Cascade control [15] and whole-body control via hierarchical inverse dynamics or task-priority quadratic programs [16]–[18] use fixed-gain nested or strictly prioritized loops without conditional gates; gain scheduling [19], [20] varies parameters continuously with one scheduling variable rather than responding to qualitatively distinct physical regimes. We additionally benchmarked a task-priority QP whole-body controller both standalone and as an assist layer on ACC (§V-E); it fails to resolve the precision–recovery trade-off because its fixed task-priority formulation lacks conditional regime-gating. Split-range, selector, and override architectures [21], [22] already implement continuous blending in process control. ACC adopts this principle; its contribution is the synthesis of physically-motivated gating surfaces (spatial proximity, temporal velocity envelope, pitch safety-interlock) derived from wheeled-biped failure modes rather than generic process-variable limits. ACC's conditional gates connect to hybrid/switched systems [23], where continuous dynamics are partitioned by discrete switching surfaces, and to hysteresis-based switching logic [24], which deliberately breaks the symmetry of a switching surface to prevent chattering—the same asymmetry ACC realizes in continuous time through its fast-attack/slow-release envelope. Smoothstep softens these surfaces to C^1 transitions. A recent review [25] surveys wheel-legged biped platforms and control approaches, noting the absence of a unified classical solution.

B. Disturbance Rejection, Anti-Windup, and ACC's Distinction

DOB [26], [27] and ADRC [28], [29] provide temporal disturbance estimation but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Classical anti-windup—back-calculation, conditional integration, and the unified observer-based framework [30]–[32]—triggers on *actuator saturation*; ACC's anchor uses **spatial, physics-conditioned gating**: the proximity gate confines integration to a 15 cm neighborhood, and the asymmetric envelope distinguishes quiet stance from ringdown via velocity envelope—replacing the estimation-speed-vs-noise trade-off with a time-domain attack-vs-release trade-off. Push recovery via capture-point [33], [34] and flight-phase control [35] is complementary. Blind stair climbing via RL was shown on Ascento [36]. Terrain adaptation uses exteroceptive elevation mapping [37], [38], learned policies that fuse proprioception with terrain estimates [39], [40], or purely proprioceptive contact sensing [41]–[43]; ACC extends proprioceptive contact-point estimation to wheeled bipeds. Hyon and Cheng [44] achieved full-body humanoid balancing by gravity compensation with optimal contact-force distribution, making the robot passive to external forces without measuring them. ACC shares the requirement of reacting to unmeasured pushes but meets it by switching regime on proprioceptive gating variables rather than by redistributing contact forces.

III. Robot Platform

A. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo [45] simulation at 500 Hz ($\Delta t_{\text{sim}} = 0.002$ s). ACC and every classical baseline: 100 Hz ($\Delta t_{\text{ctrl}} = 0.01$ s, 5 physics substeps); the classical baselines are additionally measured at their 50 Hz design rate (10 physics substeps) in §V-D1, and the PPO reference is reported at the 50 Hz rate it was trained at. All experiments use 400 Nm/s torque rate limit, raw torque commands direct to actuators. MuJoCo integrator: implicitfast (4 iterations, 8 linesearch iterations); cone: pyramidal (elliptic for wheels, μ_s=1.2, μ_k=0.8 body); armature: 0.008 kg·m^2 (wheels), 0.02 kg·m^2 (leg joints); contact stiffness/damping: `solref`={0.002,1} (body), {0.005,0.7} (wheels). The stiff body contact (`solref`={0.002,1}) idealizes a rigid ground. Softening `solref[0]` to 0.02 (tire-on-asphalt) increases planar idle RMS by ~60 % to ~1.2 mm, so every sub-millimeter idle figure reported here is conditional on stiff ground contact; tire compliance adds 0.1 – 0.5 mm on real surfaces (see §VI).

Height convention: Two different vertical references appear in this paper and are easily confused, so we fix them here and mark every height accordingly. **CoM-z** is the height of the whole-body center of mass above ground and is the quantity ACC commands and regulates; the nominal standing pose is h_{CoM}=0.404 m. **Base-z** is the torso root-body height, `qpos[2]` in the MuJoCo model; the same nominal pose has z_{base}=0.536 m, i.e. 13.2 cm above the CoM. The offset is pose-dependent but varies by less than 1 cm over the postures used here, so the two conventions differ by roughly a constant. All ACC results are reported in CoM-z. The PPO reference (§V-D) commands base-z over 0.40 – 0.70 m, and the WBC baselines (§V-E) inherit the same base-z convention; those numbers are labelled base-z where they appear and must not be compared numerically against CoM-z heights. Confusing the two is not hypothetical: it is exactly bug F3 in our own WBC implementation audit (Table XIV), where base-z was passed as a CoM-z command and produced a 13.2 cm reference error.

ACC's posture map is calibrated at two setups, at CoM heights of 0.354 m and 0.454 m (±5 cm about nominal), and interpolated between them. Outside that band the joint targets are linearly extrapolated, bounded to the interval [0.254, 0.554] m, i.e. 10 cm beyond the calibrated band at each end. This ±10 cm extrapolation bound, and not a ±5 cm one, is what the per-leg height split (§IV-F) operates against.

IV. ACC Formulation

A. Two-Channel Torque Assembly

ACC assembles a PD balance core, proximity-gated anchor integral, and posture regulation into two torque channels—wheel torque ($\boldsymbol{\tau}_w \in \mathbb{R}^2$) and leg-joint torque ($\boldsymbol{\tau}_q \in \mathbb{R}^8$)—with flight and terrain as independent extensions (Fig. 2):

$$\begin{aligned} \boldsymbol{\tau}_w &= \boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}}\boldsymbol{\tau}_{\text{anchor}} + g_{\text{flight}}\boldsymbol{\tau}_{\text{flight}}, \\ \boldsymbol{\tau}_q &= \boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}}\Delta\boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd},L},\ h^{\text{cmd},R}). \end{aligned} \tag{1}$$

Every gate g_* except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$\text{ss}(x; a, b) = 3t^2 - 2t^3, \qquad t = \text{clip}\left(\frac{x-a}{b-a},\ 0,\ 1\right). \tag{2}$$

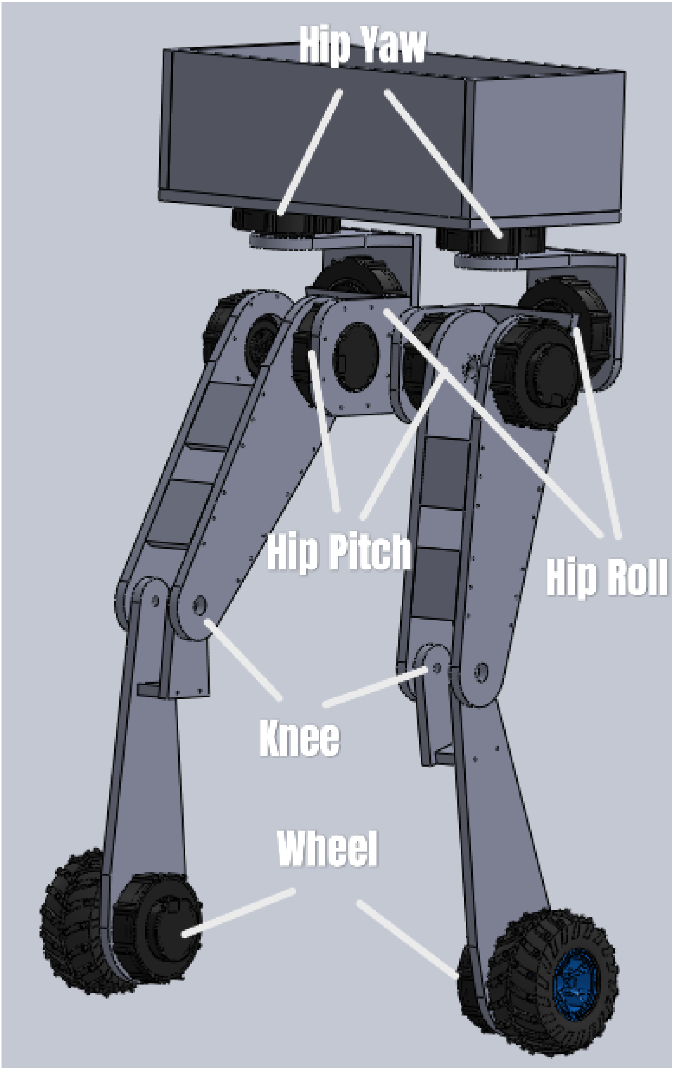

Fig. 1. Robot joint layout with world coordinate frame (right-handed, Z up). In the model as built, both wheel axles lie along world X and the two wheel links are separated by 0.271 m along X; the rolling—and therefore sagittal—direction is world Y, and world X is the lateral (track) axis. All axis-resolved results in this paper follow that convention, verified four independent ways in Sec. V-A. The CoM is located at the torso geometric center. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

Equation (2) is the standard Hermite blend: it is C^1 in x, with ss = 0 for $x \leq a$, ss = 1 for $x \geq b$, and zero derivative at both knees. Continuity matters here for a concrete reason. A hard threshold on any of these conditions injects a torque step whose magnitude equals the full gated component; at 100 Hz with a 400 Nm/s rate limit, such a step is spread over several control cycles by the rate limiter and appears to the plant as an unmodelled impulse. Smoothstep removes the step entirely: the gated component enters and leaves the torque sum with continuous first derivative, so the rate limiter is never the active constraint during a gate transition. The one exception is g_{flight}, which is a genuinely discrete load-based hysteresis with asymmetric engage/release timers (2-step engage, 5-step release); contact loss is a binary physical event, and the asymmetric timers exist specifically to prevent balance torque from re-engaging during landing bounce. Note that the composition $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); \cdot)$ is only C^0 *in time*, because the asymmetric EMA of Eq. (14) switches coefficients when $|v_t|$ crosses EMA_{t-1}; the smoothstep is still C^1 in its argument, so the resulting torque is continuous but its time derivative may jump at attack/release transitions.

Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation. The two channels are coupled only through the plant, not through the controller: no term in $\boldsymbol{\tau}_w$ reads a leg-posture state and no term in $\boldsymbol{\tau}_q$ reads the sagittal balance state. This is what allows flight recovery and terrain adaptation to be added as independent extensions (§IV-E, §IV-F) without retuning the anchor.

Relation to existing gated architectures. "Conditionally-gated" here denotes smoothstep activation by *physical* conditions (g_{prox}, g_{env}, g_θ, g_{terrain}). Continuous blending is not itself novel—split-range, selector, and override controllers have used it in process control for decades [21], [22]. Three distinctions are worth stating precisely. First, industrial override architectures select *among parallel controllers* using process-variable limits; ACC gates *components within a single torque assembly* using physical regime. Second, the gating variables are not control errors but physical quantities of three different kinds—spatial (distance from home), temporal (velocity envelope), and safety-interlock (attitude)—each chosen because a specific measured failure mode of this platform is expressible in it (§VI). Third, ACC is not cascade control [15]: there is no inner/outer loop hierarchy and no setpoint is passed between controllers. The contribution is therefore the *synthesis*—which gating surfaces, on which variables, with which asymmetry—not the blending mechanism.

B. Sagittal Balance Core ($\boldsymbol{\tau}_{balance}$)

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\boldsymbol{\tau}_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} &= \tau_{\text{lat}} + \tau_{\text{yaw}}. \end{aligned} \tag{3}$$

where τ_{com} and τ_{diff} are common-mode and differential components, respectively. The individual terms are:

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \tag{4}$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \tag{5}$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10\,\text{m}), \tag{6}$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \tag{7}$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_\psi + k_{d,\text{yaw}} \cdot \omega_z, \tag{8}$$

where θ is the torso pitch angle, $\dot{\theta}$ is the pitch rate, v_{sag} is the sagittal body velocity, Δx is the sagittal position error from the latched home position, ϕ is the roll angle, and e_ψ is the yaw error.

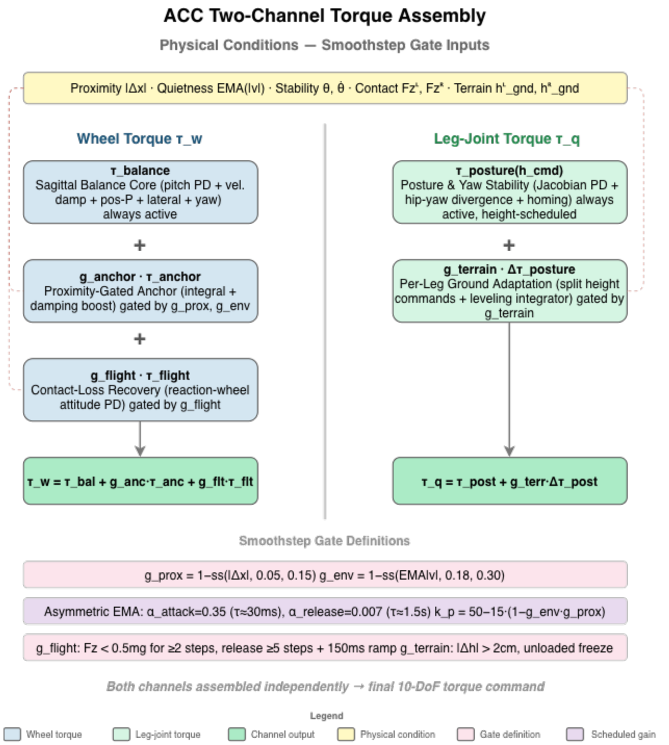

Fig. 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

P-only limitation. The P position term (k_{pos}, clipped to ± 0.10 m, saturating at ± 4 Nm) keeps the robot within ± 4–6 cm of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle—motivating the anchor integral.

Why saturation-triggered anti-windup is insufficient. Saturation-triggered AW triggers on actuator saturation; on this platform the critical failure is **state-dependent**: during a push, position error grows to 10–30 cm while wheels remain within their 20 rad/s range, so saturation-triggered AW never activates. ACC's proximity gate provides spatial confinement via continuous smoothstep—avoiding both over-restriction (blocking slow-drift correction) and under-restriction (allowing windup during sustained pushes). The simplest conditional-integration AW is evaluated in Appendix A; broader AW families remain open.

Pitch stiffness. Pitch stiffness is fixed at k_p=50 in the canonical ACC architecture. The anchor and boost alone resolve the precision-push trade-off without gain scheduling; a scheduled variant that softened k_p from 50 to 35 during pushes was tested and rejected—it did not improve push survival (S0 vs. S1, Table IV) and is excluded from the canonical specification.

C. Proximity-Gated Anchor Mechanism ($g_{anchor} \cdot \tau_{anchor}$)

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \quad (9)$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is "at home and quiet" and transition smoothly to 0 otherwise:

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}(|\Delta x|;\ 0.05\,\text{m},\ 0.15\,\text{m}), \quad (10)$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|);\ 0.25\,\text{m/s},\ 0.50\,\text{m/s}), \quad (11)$$

$$g_\theta(\theta, \dot{\theta}) = \left[1 - \text{ss}(|\theta|;\ 2^\circ,\ 12^\circ)\right] \times \left[1 - \text{ss}(|\dot{\theta}|;\ 2^\circ/\text{s},\ 15^\circ/\text{s})\right], \quad (12)$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_\theta, \quad (13)$$

where $\text{ss}(x; a, b)$ denotes the smoothstep of Eq. (2) with lower and upper knees a and b. Each gate is therefore a dimensionless, monotone, C^1 function that equals 1 inside its "home and quiet" band, falls smoothly to 0 outside it, and never switches discontinuously. Three physically distinct conditions must hold simultaneously for the anchor to integrate: the robot must be spatially near home (g_{prox}), temporally quiet (g_{env}), and attitudinally settled (g_θ).

The pitch stability gate g_θ closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ($|\theta| < 2^\circ$, $|\dot{\theta}| < 2^\circ/\text{s}$), $g_\theta = 1$ and the boost is fully available.

Asymmetric envelope follower. The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1-\alpha_a)\text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1-\alpha_r)\text{EMA}_{t-1}, & \text{otherwise}, \end{cases} \quad (14)$$

with time constants $\tau_{\text{attack}} \approx 23$ ms and $\tau_{\text{release}} \approx 1.5$ s as the primary specification; the discrete coefficients are derived via $\alpha_a = 1 - \exp(-\Delta t/\tau_{\text{attack}}) \approx 0.35$ and $\alpha_r = 1 - \exp(-\Delta t/\tau_{\text{release}}) \approx 0.007$ at 100 Hz (Δt=0.01 s). This asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds the EMA in a middle band where the gate is half-open and the boost cannot collapse the cycle (validated by the symmetric-EMA ablation, Section V). The fast-attack/slow-release envelope jumps to 1 only when oscillation truly decays, and drops to 0 within ~ 25 ms of a disturbance. The mechanism is a peak-hold/envelope detector with asymmetric time constants; ACC's contribution is coupling envelope detection to a spatial proximity gate for integral conditioning. Fig. 3 illustrates the complete gate activation sequence during a push-recovery cycle.

EMA initialization. EMA_0=0 biases $g_{\text{env}} \approx 1$ for $\sim$7.5 s at startup, decaying as velocity noise accumulates. Measurements use 3–5 s settle (see Table captions); hardware should initialize EMA_0=1.0.

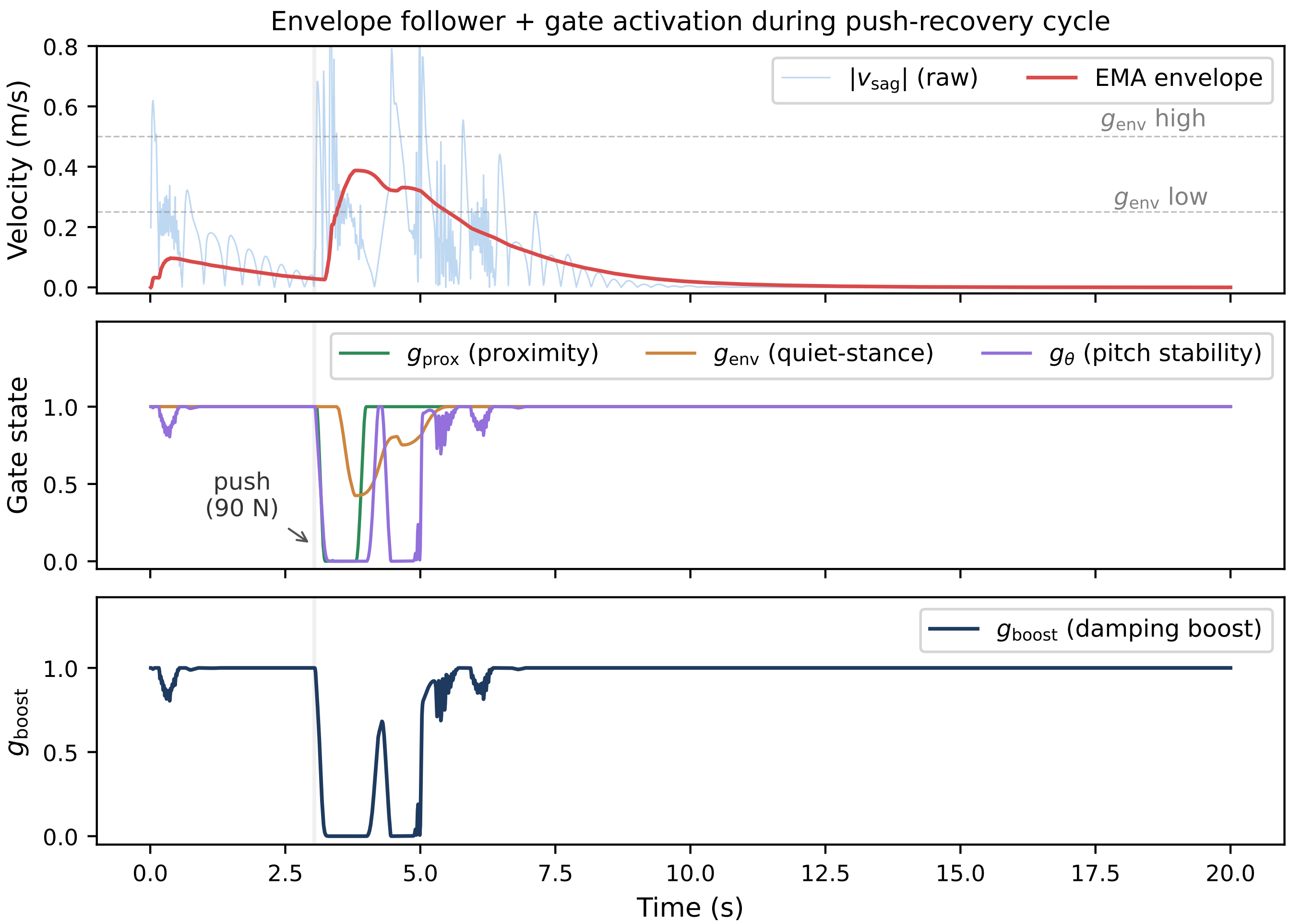

Fig. 3. Gate activation during a 90 N lateral (world $+x$) push-recovery cycle (100 Hz control). **Top:** $|v_{sag}|$ and asymmetric EMA envelope; dashed lines mark the 0.25 – 0.50 m/s quiet-stance band. **Middle:** Individual gate states. **Bottom:** Composite damping boost g_{boost}. Grey shading: 7-step push window. The three panels are the empirical basis for finding (iii) of §V-A4: g_{prox} and g_{env} both sit pinned at their quiet-stance values before the impulse and after the ringdown, which is why ablating either reproduces ACC bit-for-bit at idle, and why only a disturbance experiment can measure their contribution.

D. Posture and Yaw Stability ($\boldsymbol{\tau}_{posture}$)

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\boldsymbol{\tau}_{\text{posture}} = \boldsymbol{\tau}^{\text{jac}}_{\text{PD}}(q, \dot{q}, h^{\text{cmd}}) + \boldsymbol{\tau}_{\text{div}}(q^L_{\text{hy}}, q^R_{\text{hy}}) + g_{\text{settled}}\, \boldsymbol{\tau}_{\text{homing}}(q - q_{\text{ref}}), \tag{15}$$

where $\boldsymbol{\tau}_{\text{div}}$ damps hip-yaw divergence and $\boldsymbol{\tau}_{\text{homing}}$ restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

E. Contact-Loss Recovery ($g_{flight} \cdot \boldsymbol{\tau}_{flight}$)

When both wheels lose ground contact, ACC detects this via per-wheel normal forces and substitutes a flight attitude controller using the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F^L_z, F^R_z < 0.5mg \text{ for } \geq 2 \text{ steps}, \\ 0, & \text{if } \min(F^L_z, F^R_z) \geq 0.5mg \text{ for } \geq 5 \text{ steps}, \end{cases} \tag{16}$$

$$\boldsymbol{\tau}_{\text{flight}} = \text{clip}\big(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\text{wheel}},\ \pm 2\,\text{Nm}\big). \tag{17}$$

Forward wheel spin pitches nose-up during descent; 5-step release hysteresis prevents balance-torque re-engagement during bounce. The free-fall drop is a stress test and the ledge drive-off the intended use case. We note in advance of the results that this override is the one ACC mechanism whose ablation does *not* show it to be necessary at the reported scales: the wheel–inverted-pendulum loop it replaces is itself a reaction-flywheel controller when the wheels are unloaded, and the override's measured contribution is to bound landing-attitude variance near the upper end of the drop envelope rather than to enable recovery (§V-C, Table V).

F. Per-Leg Ground Adaptation ($g_{terrain}$)

When the two wheels rest at different ground heights—a curb straddle, an oblique ledge—a single torso height command forces the torso to roll by the step angle. ACC instead maintains an *independent ground estimate per leg* and issues two height commands. The estimate is taken from the wheel contact-point height, which is exact at any body tilt for a rigid wheel, and is updated only while that wheel carries real load:

$$\hat{h}^i_{\text{gnd}} \leftarrow \hat{h}^i_{\text{gnd}} + \alpha_g \left(z^i_{\text{contact}} - \hat{h}^i_{\text{gnd}}\right), \quad F^i_z \geq 0.2mg, \quad i \in \{L, R\}, \tag{18}$$

with $\alpha_g = 0.35$ per control step to reject contact chatter. The $0.2mg$ load test rather than mere geometric touching is deliberate: a lightly touching wheel that is in the act of unloading rides a few centimetres high and injected a 2.9 cm phantom ground step on the 15 cm curb.

An unloaded wheel does *not* update (18). It freezes its estimate, because an airborne wheel supplies no information about where its ground is. The one exception is the only physically unambiguous "my ground is gone" signal—the wheel centre descending *below* its own believed ground:

$$\delta^i > 3\,\text{mm} \;\Rightarrow\; \dot{\hat{h}}^i_{\text{gnd}} = -\left(v_0 + k_s\,\delta^i\right), \tag{19}$$

where $\delta^i = \hat{h}^i_{\text{gnd}} - z^i_{\text{contact}}$ is the depth by which the wheel has fallen below its believed ground, with $v_0 = 0.5\,\text{m/s}$ and $k_s = 50\,\text{s}^{-1}$. The proportional term matters: a fixed slew converged the ground estimate in 0.30 s, by which time a 20 cm dismount had already tipped the robot past recovery, whereas a 15 cm drop under (19) slews at 8 m/s and converges within two control steps, so the leg extends *during* the fall rather than after it. When both wheels are airborne, both estimates track the mean wheel height, which reproduces the symmetric pre-adapter behaviour exactly and leaves ballistic flight to §IV-E.

The step $\Delta h = \hat{h}^L_{\text{gnd}} - \hat{h}^R_{\text{gnd}}$ is then absorbed by the legs rather than the torso. The split is kinematic, not a linear height offset: with $L(\cdot)$ the forward-kinematic leg length and L^{-1} its inverse over the calibrated posture range, the leg on the higher ground is shortened by the full step,

$$L_{\text{high}} = L\left(h^{\text{cmd}}\right) - |\Delta h|, \tag{20}$$

$$h^{\text{high}} = L^{-1}\left(L_{\text{high}}\right), \qquad h^{\text{low}} = h^{\text{cmd}}, \tag{21}$$

both clamped to the calibrated envelope extended by 10 cm at each end. To first order this is the familiar $h^{\text{cmd}} \mp \Delta h/2$ symmetric split, but the $h \mapsto L$ map is markedly nonlinear near full squat: the linear proxy erred by 12 cm at the lower extrapolation limit, folding the knee back under the thigh. If L_{high} falls below the shortest physically reachable leg, ACC raises the *whole* robot until the short leg fits, trading commanded height for the ability to span the step. A 20 cm curb at nominal stance exceeds even this: it would require 170° of knee flexion against a 2.7 rad limit, and the uncompensated remainder appears as a predicted torso roll $\arctan(-\varepsilon/w_{\text{track}})$ with ε the unspanned step and $w_{\text{track}} = 0.278\,\text{m}$ the wheel separation. Reporting this expected roll explicitly is what keeps a legitimate straddle from being misread as an incipient fall by the tilt-based safety monitors.

TABLE I
CAPABILITY COMPARISON ACROSS WHEELED BIPED PLATFORMS. "—" = NOT REPORTED IN CITED WORK. IDLE PRECISION AND PUSH FORCE ARE NOT STANDARDIZED BENCHMARKS; QUANTITATIVE CROSS-PLATFORM COMPARISON IS LIMITED BY METRIC HETEROGENEITY.

Metric	Ascento	DIABLO	Whleaper	SKATER	**ACC (Ours)**
Balance	LQR	LQR + aux. loops	LQR / PPO (per mode)	Hierarch. MPC	**Gated PD + anchor I**
Idle precision	—	—	—	—	**0.294** mm **sag. CoM RMS**
Push recovery	—[a]	—	—	—	**8-dir.,** F_{med} = 117 N
Height	Discrete jump	Stance[b]	Multi-modal	Contin.	**0.35** – 0.45 m CoM[c]
Flight	Jumping (airtime)	—	—	—	**Drop 100** cm, **ledge 50** cm
Terrain	—	—	—	Vision-based	**Per-leg contact**
Validation	Hardware	Hardware	Hardware	Hardware	**Simulation**

[a]Qual. omnidirectional push recovery demonstrated in supplementary video; quantitative force thresholds not reported in cited work. [b]Height variation demonstrated; quantitative range not reported.
[c]Calibrated CoM-z band (§III-A), ±5 cm about the 0.404 m nominal, with joint targets extrapolated to 0.254 – 0.554 m; commanded ±5 cm transitions are evaluated under disturbance in §V-F. Not comparable to the base-z ranges of the cited platforms or of our PPO reference.

Because the $h \mapsto L$ table is itself extrapolated at the extremes, a purely feedforward split left a systematic 6° torso roll on the 10 cm curb. A closed-loop leveling integrator trims the split by the *measured* roll at 0.15 m/(rad s) with ±6 cm of authority. It integrates only when all three of the following hold: both wheels loaded, a genuine ground step $|\Delta h| \geq 2\,\text{cm}$, and a ground estimate settling at less than 0.15 m/s. The gating is not cosmetic. Without the step condition, the dynamic roll of a turn is integrated as if it were terrain and wound an 8.6° roll bias through a 180° turn; without the rate condition, the geometric transient of a curb climb is integrated as a leveling error and compresses the flat leg. Off-step, the trim decays to zero with a 1 s time constant.

G. Complete Numerical Parameter Set

Table II lists core parameters; the full set (50+ gains) is available in the supplementary material.

V. EXPERIMENTAL VALIDATION

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. ACC and all six classical baselines are evaluated at 500 Hz simulation / 100 Hz control, so the headline comparison is rate-matched by construction. The PPO reference is reported at the 50 Hz rate its policy was trained at. All configurations use a 400 Nm/s torque rate limit per joint.

TABLE II

CORE NUMERICAL PARAMETERS FOR THE ACC BALANCE CONTROLLER. ALL VALUES IN SI UNITS UNLESS NOTED OTHERWISE.

Balance Core		
k_p	50	Nm/rad
k_d	3.0	Nm/(rad/s)
k_{pos}	20	Nm/m
Position clip	±4.0	Nm
k_{vel}	15	Nm/(m/s)
Vel. damp. scale	1.5	—
k_{roll}	55	Nm/rad
$k_{\text{roll},d}$	5.5	Nm/(rad/s)
k_{yaw}	1.0	Nm/rad
$k_{\text{yaw},d}$	0.3	Nm/(rad/s)
$k_{\text{pos,return}}$	2.0	Nm/m
k_{heading}	2.0	Nm/rad
$k_{\text{heading},d}$	0.6	Nm/(rad/s)
Anchor Mechanism		
K_i (anchor integral)	4.0	Nm/(m·s)
Integral cap	2.0	Nm
Integral leak	5×10^{-3}	per step
Boost scale k_{vb}	5.0	—
g_{prox} band	0.05–0.15	m
g_{env} band	0.25–0.50	m/s
τ_{attack}	0.023	s
τ_{release}	1.5	s
g_{stab} (θ)	2–12	°
g_{stab} ($\dot{\theta}$)	2–15	°/s
Flight / Terrain / Other		
k_{flight} (P / D)	2.0 / 0.5	—
Wheel damping (flight)	0.02	Nm/(rad/s)
Flight engage / release	2 / 5	steps
F_z thresh. (flight / terrain)	0.5 / 0.2	$\times mg$

Control rate. Every classical baseline was measured at two control rates under an otherwise identical protocol, with episode length held fixed in seconds rather than in steps: ACC's 100 Hz, which is what Table VII reports, and 50 Hz, the rate for which the gains were tuned. Both sets appear side by side in §V-D1 (Table VIII). The fall rate is 100 % in all twelve cells, so the comparison does not turn on which rate is chosen. The converse direction—ACC at 50 Hz—is a negative result: with the discrete coefficients recomputed for the longer step ($\alpha_a\approx0.58$ and $\alpha_r\approx0.013$, re-derived from Eq. (14) at Δt=0.02 s, with the anchor leak rescaled to 0.010 per step) the controller is not preserved. The robot collapses, and the mean CoM height over the recorded window is 0.124 m against a nominal standing height of 0.404 m. The 0.050 mm "idle RMS" and the 10 N push threshold produced by that run are artifacts of a robot lying motionless on the ground—10 N is the lower bound of the binary search, returned when even the smallest tested impulse fails. No 50 Hz ACC result is therefore reported, and rate-independence in the downward direction is not claimed; this is analysed as Limitation 5. RNG seeds are deterministic: idle standing uses seed = 20260727 + trial_id; push ablation uses seed = $20260728 \times 100 +$ rep; LQR baselines cycle seeds $\{0, 42, 123\}$ across episodes.

A. Anchored Standing

1) Two precision metrics, deliberately kept separate: "Standing precision" names two different quantities in the wheeled-biped literature, and they are not interchangeable. Let $\mathbf{p}(t) \in \mathbb{R}^2$ be the horizontal base position over an evaluation window of length T, let $\mathbf{p}_0$ be the commanded home position, and let $\bar{\mathbf{p}} = \frac{1}{T}\int_0^T \mathbf{p}(t)\,dt$ be the trial mean. We report

$$A = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2\,dt\right)^{1/2}, \tag{22}$$

$$R = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2\,dt\right)^{1/2}. \tag{23}$$

A is the *accuracy*: RMS displacement from the position the controller was told to hold. It retains any DC parking offset, so it answers "how close to the commanded pose does the robot actually park?". R is the *repeatability*: RMS about the trial's own mean, with the DC offset removed. It answers "how still is the robot once it has parked?". A controller with a persistent 27 mm offset and a perfectly steady hold scores A=27 mm but $R\approx0$ mm; a controller centred on the target but oscillating scores the reverse. Reporting either alone is uninformative, and—because the two differ by more than an order of magnitude on this platform—comparing a value of one against a value of the other produces a meaningless ratio.

Both are reported for every configuration below, resolved separately along the sagittal axis, the lateral axis, and the full plane, because the two horizontal axes of this platform are not merely different in degree—they are actuated by different means and must not be pooled or interchanged. We therefore fix the convention explicitly, since the two are easy to transpose and the physical conclusions invert if they are:

- The **sagittal** axis is world y. Both wheel axles point along world x and the two wheels are separated along world x (w_{track}=0.278 m), so the rolling direction—the axis on which wheel torque produces ground force, and the only axis with a genuine inverted-pendulum mode—is world y. The controller agrees: `init_v3_controller` packs $\hat{\mathbf{s}} = (0, 1)$ as its sagittal unit vector.
- The **lateral** axis is world x. No actuator produces lateral ground force; the axis is held passively by the wheel contacts and by the roll restoring moment of the track width.

Two dynamic checks confirm the assignment rather than merely asserting it. A 90 N impulse along y is absorbed with a peak wheel speed of 52 rad/s and leaves a residual offset of 4.8 mm—the wheels drive back under the CoM—whereas the same impulse along x leaves a permanent 23.2 mm offset at a peak wheel speed of 18 rad/s, because nothing can push the robot sideways again. And over a 20 s quiet stance the robot rolls 24.6 mm along y with the wheel midpoint tracking the CoM to within 0.2 mm, while x holds to 0.02 mm. All four checks are reproduced by `scripts/verify_body_axes.py`, which asserts on disagreement.

This matters for reading Table III: the anchor integral and the gated damping boost both live in the wheel-torque channel, so *both act on the sagittal axis only*. The lateral column is therefore a measure of how little the plant moves when nothing drives it, not a measure of controller quality, and the sagittal and planar columns carry essentially all of the information about what ACC does (Sec. VI, Limitation 8).

2) Unified measurement protocol: Every row of Table III was re-measured under one protocol so that no two entries in the same column come from different definitions. Each trial runs 25 s at 500 Hz physics / 100 Hz control from the nominal standing pose with randomized initial conditions (σ=0.005 rad per joint, clipped to joint limits; σ=1 mm on root height), seeded seed=20260727+trial_id for trial_id $\in$ $[0, 10)$. The first 5 s are discarded as settling—long enough to cover the EMA_0=0 transient (~7.5 s to full release, but <5 s to within 1 % of the quiet-stance value)—and A and R are computed over the remaining 20 s. Displacements are taken relative to each trial's own t=0 pose, so $\mathbf{p}_0$ is the commanded home rather than the world origin. A trial is scored as a fall if $|\theta|$>46° or root height <0.15 m at any point. Confidence intervals are Student-t ($t_{9,0.025}$=2.262, SEM=SD/$\sqrt{10}$).

The protocol has one clear scope limit: because every trial starts at the commanded home, it never exercises the anchor's return-to-home function. It measures the controller's ability to *stay* put, not to *come back*. The latter is measured by the push experiments of Sec. V-B. This distinction turns out to matter for attributing the mechanism (Sec. V-A4).

3) Result: ACC holds **sub-millimeter sagittal repeatability**, R_{sag} = 0.294±0.015 mm (95% CI $[0.283, 0.305]$ mm), across ten randomized trials with no falls. This is the figure of merit for the platform, because sagittal is the actuated inverted-pendulum axis: it is the axis on which the robot can fall, the axis the wheel-torque channel drives, and the axis that every configuration in the ladder is trying to hold.

Signal-to-noise ratios against a simulator noise floor are omitted here. The floor is obtained by locking the joints and pinning the base (`run_r2_between_trial_stats.py--noise-floor`); because that procedure zeroes velocities without welding the base, it is trustworthy on the laterally-constrained axis (0.0019 mm RMS) but introduces artificial sag on the gravity-loaded sagittal and vertical axes, where the residuals it reports (5.79 mm and 63.5 mm) reflect the pinning rather than integrator noise. Precision is therefore assessed from the relative spread *between* configurations measured under an identical protocol: removing a single mechanism (S2) moves R_{sag} from 0.294 to 4.772 mm, a factor of 16 with non-overlapping SDs, which establishes that the metric resolves the effect under study.

Against the P-only baseline (L0), which sustains the 69.7 mm peak-to-peak sagittal limit cycle described in Sec. I, the like-for-like improvements are

- sagittal repeatability R_{sag}: 17.461 → 0.294 mm, a factor of **59.4**;
- planar repeatability R_{xy}: 17.845 → 0.788 mm, a factor of **22.6**;
- lateral repeatability R_{lat}: 3.675 → 0.730 mm, a factor of **5.0**.

The ordering of these three factors is itself informative and worth stating plainly, because it is the opposite of what a reader might expect. The largest gain is on the sagittal axis, which is the hard axis; the smallest is on the lateral axis, which is the easy one. That is the correct signature for a wheel-torque controller: every term ACC adds beyond L0—position homing, the anchor integral, the gated damping boost—produces ground force along the rolling direction and none of them can produce lateral force at all. The residual 5× lateral improvement is an indirect consequence of the robot pitching and yawing less rather than of any lateral control action. A reader comparing against platforms that report a single scalar "standing precision" should compare against R_{sag}, not against R_{lat} or a planar norm, since the latter two are diluted by an axis this class of platform does not actuate.

The accuracy picture is markedly worse than the repeatability picture, and on the same axis. ACC parks $\bar{s}$ = −27.0 mm from the commanded home along the sagittal axis and holds that offset to within 0.009 mm SD across trials—the robot rolls backwards by 27 mm during the settle and then stands there, exceptionally still, in the wrong place. Planar accuracy is therefore A_{xy} = 26.994 mm and improves only 1.8× over L0 (49.1 → 27.0 mm), a full order of magnitude worse than the repeatability gain. The offset is not a measurement artifact and not an unregulated axis: it lies precisely on the axis the anchor integral does control, and the integral reduces it only from 31.2 mm (M1, K_i=0) to 27.0 mm. An ideal integrator would drive it to zero. We report the discrepancy as the principal open weakness of the present design (Limitation 8) rather than absorbing it into a planar average where it is less visible.

4) Which mechanism produces the precision: The bottom two blocks of Table III isolate the contributions. Four findings are worth stating precisely, because three of them contradict the attribution that the gate structure invites.

(i) Idle steadiness comes from the damping ladder; the integral buys accuracy instead. The two contributions separate cleanly once the axes are kept apart, and they do not compete for the same metric.

M1 disables the anchor integral entirely (K_i=0) while leaving every gate in place, and its *repeatability* is not merely as good as ACC's but slightly better (R_{sag} 0.089 mm vs. ACC's 0.294 mm; R_{xy} 0.733 mm vs. 0.788 mm). Its *accuracy*, however, is worse by 4.2 mm: it parks at $\bar{s}$ = −31.2 mm against ACC's −27.0 mm. That is the integral's entire measurable idle signature, and it is exactly the trade one expects from an integral term—it walks the robot closer to the commanded home at the cost of a small amount of steadiness, since the accumulating term is itself a slowly-varying command. A protocol that reports repeatability alone will conclude, wrongly, that the anchor integral is inert at idle; a protocol that reports accuracy alone will miss that it costs a factor of three in steadiness.

Conversely M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, and

TABLE III
IDLE STANDING PRECISION ACROSS THE FULL ABLATION LADDER. ALL ROWS RE-MEASURED UNDER THE SINGLE PROTOCOL OF SEC. V-A; VALUES ARE MEAN±SD OVER N=10 TRIALS. SEE THE NOTE BELOW THE TABLE FOR COLUMN DEFINITIONS.

	Configuration	R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	A_{lat} (mm) lateral acc.	Survival
Additive ladder (bottom-up)							
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	−44.8	9.933 ± 0.712	10/10
L1	+ position homing (±0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	−30.4	2.629 ± 0.884	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	−27.5	3.661 ± 1.264	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
Proposed controller							
L3/S1	**ACC** (full anchor, fixed k_p=50)	**0.294 ± 0.015**	**0.788 ± 0.069**	**0.730 ± 0.073**	**−27.0**	**1.213 ± 0.554**	**10/10**
Subtractive ablation (top-down from ACC)							
S0	ACC + scheduled k_p (rejected variant)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S2	− gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	−26.8	1.260 ± 0.616	10/10
S3	− asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S4	− proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S5	− pitch safety gate g_θ	—	—	—	—	—	**0/10**
Single-mechanism isolation (ACC with one term disabled)							
M1	K_i=0 (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	−31.2	1.192 ± 0.535	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
M3	K_i=0 *and* boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	−31.1	1.241 ± 0.591	10/10

Protocol. N=10 trials, 100 Hz control, 20 s measurement window after a 5 s settle, clean sensors, 0 ms actuator delay.
Columns. R = repeatability (RMS about the trial mean, Eq. (23)); A = accuracy (RMS from commanded home, Eq. (22)); $\bar{s}$ = mean signed sagittal parking offset. Sagittal is world y and lateral is world x (Sec. V-A). The anchor integral and the damping boost act on the sagittal axis only, so R_{sag} and $\bar{s}$ are the columns that discriminate between configurations, while R_{lat} measures a passively held axis and is near-constant by construction.
Omitted columns. A_{sag} and A_{xy} carry no information beyond the columns shown: over all twelve surviving rows the identities $A_{\text{sag}} = \sqrt{\bar{s}^2 + R_{\text{sag}}^2}$ and $A_{xy} = \sqrt{A_{\text{sag}}^2 + A_{\text{lat}}^2}$ hold to 0.003 mm and 0.02 mm respectively—accuracy on an axis is just that axis's parking offset and its repeatability added in quadrature.
Produced by `scripts/collect_idle_ladder.py` (per-trial worker `scripts/_idle_ladder_worker.py`); raw output `outputs/paper_verification/idle_ladder.json`.

reproduces the L2/L2.5 limit cycle exactly (R_{sag} = 25.198 mm). Steadiness is therefore attributable to the damping terms, in two stages: raising the velocity-damping scale takes R_{sag} from 25.2 mm to 3.97 mm (M3, boost still off), and enabling the gated damping boost takes it from 3.97 mm to 0.294 mm (ACC)—a combined factor of 86 decomposed into factors of ∼6.3 and ∼13.5. S2 confirms the second stage independently from the top down: removing only the boost degrades R_{sag} from 0.294 mm to 4.772 mm.

(ii) The damping boost acts sagittally, as it must. S2 degrades sagittal repeatability by a factor of 16 (0.294 → 4.772 mm) while leaving lateral repeatability statistically unchanged (R_{lat} = 0.703 mm vs. ACC's 0.730 mm, overlapping CIs). This is the only outcome consistent with the architecture: the boost multiplies a velocity-damping coefficient inside the wheel-torque channel (Eq. (3)), and wheel torque generates ground force along the rolling direction only. The residual oscillation it suppresses is therefore a sagittal pitching/rolling-forward mode, and an evaluation that reported the lateral axis alone would conclude, wrongly, that the boost does nothing at idle. The distinction matters because the lateral column is the one that reads as a plausible "x-axis RMS" to report by default, yet it is the one column in Table III that is nearly blind to every mechanism in the ladder: across all twelve surviving configurations R_{lat} spans only 0.70 – 3.68 mm, and across the seven rows that retain the full damping ladder (S0–S4, M1, M3) only 0.70 – 0.73 mm.

(iii) The proximity gate and the asymmetric envelope are inert during quiet stance. S4 (proximity gate removed) and S3 (symmetric envelope) are *bit-identical* to ACC on every idle metric. This is the expected behavior rather than a null result: during quiet stance the base never leaves the 5 cm inner band, so $g_{\text{prox}} \equiv 1$ and gating it changes nothing; and the velocity envelope never rises into the attack branch, so the asymmetric and symmetric followers produce the same signal. Both mechanisms are defined by what they do when the robot is *displaced*—windup prevention during a push (S4) and ringdown after one (S3)—and are correctly evaluated only in Table IV. The idle protocol, which starts at home and stays there, is blind to both by construction.

(iv) The pitch safety gate is required even to stand. S5 falls in 10/10 trials during pure quiet stance, with no push applied. Removing g_θ leaves the damping boost free to act on the pitch state at large excursions, and the resulting positive feedback diverges from the ±0.005 rad initial perturbation alone. This is a stronger safety result than the push-only evidence previously reported for this ablation.

Taken together: the anchor's *gates* are what make an aggressive damping and integral configuration admissible without windup or divergence, while the damping terms they admit are what deliver the steadiness. The gates are enabling conditions, not the proximate cause of the precision—a distinction the earlier framing of this work conflated. The integral's primary role is drift rejection under disturbance, which by construction cannot appear in a protocol that starts at the target; its measurable idle signature is confined to the 4.2 mm reduction in sagittal parking offset noted above (M1 −31.2 vs. ACC −27.0 mm), bought at a 3× cost in sagittal repeatability. Both effects sit on the sagittal axis, and neither is visible on the lateral one.

Post-push velocity ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle. The symmetric-EMA ablation exhibits bistable failure: a post-push cycle holds the envelope at ≈0.10, leaving the boost gate half-open, and the robot oscillates perpetually. The ungated-integral ablation winds up during the push and prevents recovery entirely. All three of these are post-push behaviors and are quantified in Sec. V-B.

B. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: 10 − 160 N impulses (7 control steps) at the torso in 8 directions. Bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so $\theta = \pm 90°$ are the sagittal (fore/aft) bearings and $\theta = 0°/180°$ the lateral ones, per the axis convention of Sec. V-A. $F_{\min}$ is the maximum magnitude survived in *every* direction; F_{med} is the median. All thresholds measured via binary search over $[10\,\text{N}, 160\,\text{N}]$ (8 iterations per direction, 5 N tolerance). Survival: torso pitch $< 46°$ for 17 s post-push.

Single-bearing stress tests. Several auxiliary experiments in this paper—the robustness sweep (Table X), the rate-limit sweep, the ringdown and gate-activation traces (Figs. 3, 5), and the compound-disturbance protocol (Table XI)—apply a single impulse along world $+x$ rather than repeating the full 8-bearing sweep, which would multiply their cost eightfold. That bearing is *lateral*, and it is a below-median one: the polar sweep places 0° at 94.3 N, third-weakest of the eight and 22.6 N below F_{med}. Those experiments are therefore conservative probes of ACC's disturbance margin, not optimistic ones, and we label them as lateral throughout rather than as "forward." The omnidirectional figures ($F_{\min}$, F_{med}) are the ones that characterize the full envelope.

Table IV reports the factorial ablation. Additively: L1 (homing) adds 4.8 N over L0 ($p<0.001$); L2 (global integral) adds only 1.3 N over L1 (n.s.); L3 (full anchor) adds 6.7 N over L2 ($p<0.001$). The L2→L3 step bundles four mechanisms at once, and the push metrics alone cannot separate them; Sec. V-A4 performs that separation on the idle side. What the push data do show is that the proximity gate contributes nothing to push *margin* on its own (L2→L2.5, −0.6 N, n.s.), consistent with its role being windup prevention rather than capture-basin enlargement, and that the full anchor recovers a margin advantage over the ungated integral (+6.7 N) while simultaneously being the only configuration in the ladder that settles. Subtractively: S1 (fixed k_p=50, canonical) matches or exceeds S0 in all metrics, confirming gain scheduling is unnecessary—the anchor and boost alone resolve the trade-off. The gated damping boost (S2) is the dominant idle-precision mechanism, and it acts sagittally: removing it degrades sagittal repeatability sixteen-fold (0.294 mm→4.772 mm) while leaving lateral repeatability statistically unchanged (0.703 mm vs. ACC's 0.730 mm, Table III). Its push margin is, if anything, slightly better without the boost ($F_{\min}$ = 84.2 N vs. 82.8 N), so the boost buys sagittal steadiness and settling, not capture basin. S5 (pitch gate) is critical to safety—removing it collapses $F_{\min}$ to 8.8±0.6 N (~9× below ACC) and additionally

TABLE IV
FACTORIAL ABLATION OF ACC COMPONENTS: PUSH-RECOVERY AND RINGDOWN METRICS. CLEAN SENSORS, ALL ROWS N=10, MEAN±STD. SEE THE NOTE BELOW THE TABLE FOR DEFINITIONS AND SIGNIFICANCE TESTS.

	Configuration	$F_{\min}$ (N)	F_{med} (N)	Ringdown
Additive (bottom-up)				
L0	P-only (PD, no pos., no I)	70.0±1.0	122.1±2.1	never
L1	+ Pos. homing (±0.10 m)	74.8±0.8	130.5±1.2	never
L2	+ Integral K_i (no gate)	76.1±1.1	132.0±2.0	—
L2.5	+ Prox. gate only (no EMA, no boost)	75.5±0.9	132.0±2.3	—
L3	+ Full anchor (gate, EMA, boost)[7]	**82.8±1.0**	116.8±4.7	**9.0** s
Subtractive (top-down from ACC)				
S0	ACC + scheduled k_p (rejected)[6]	79.4±2.6	114.0±5.3	9.0 s
S1	**ACC**[6] (canonical, fixed k_p=50)	**82.8±1.0**	116.8±4.7	**9.0** s
S2	−Damping boost	84.2±1.1	115.5±5.3	never[1]
S3	−Asym. EMA (sym. EMA)	81.6±2.3	109.2±5.4	never[2]
S4	−Prox. gate (global I)	80.0±2.5	114.0±5.3	windup[3]
S5	−Pitch gate g_θ	8.8±0.6[5]	35.4±10.3	never[4]

$F_{\min}$ and F_{med} are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance. Idle-precision metrics for these same configurations are reported separately in Table III under a single measurement protocol; they are deliberately not duplicated here, because idle and push behaviour are governed by different subsets of the gates.
[1]Boost necessary for ringdown. [2]Symmetric EMA bistable: post-push oscillation holds EMA in mid-band. [3]Without the proximity gate the integral accumulates throughout the displacement and saturates its cap; the robot survives the impulse but does not settle. At idle this ablation is bit-identical to ACC (Table III), since the base never leaves the gate's inner band.
[4]Removing the pitch gate lets the damping boost act on the pitch state at large excursions; the resulting parametric amplification diverges. This ablation additionally fails 10/10 at pure idle with no push applied (Table III).
[5]Below measurement floor (10 N); robot cannot survive minimal push.
[6]S1 (fixed k_p=50) is the canonical ACC configuration. S0 (scheduled k_p, softened from 50 to 35 during pushes) was tested and rejected—it did not improve push survival. [7]L2→L3 = canonical ACC (S1, fixed k_p=50); bundles four mechanisms (prox. gate, asym. EMA, boost, pitch gate). S2–S5 isolates each subtractively. Welch's unpaired t-test (two-sided, N=10 per group, independent trials; Welch–Satterthwaite df∈[9, 18]; uncorrected p, Bonferroni-adj. α=0.007, see §VI): L0→L1 +4.8 N ($p<0.001$); L1→L2 +1.3 N (n.s.); L2→L2.5 −0.6 N (n.s., gate alone adds nothing); L2.5→L3 +7.3 N ($p<0.001$). S5 (pitch gate) is critical to safety ($F_{\min}$=8.8 N, and 0/10 survival even at idle); S2 and S3 are required for ringdown. Idle metrics for every row are in Table III. With N=10, 95%CI half-width ≈0.72×std ($t_{0.025,9}$=2.262, single-group df=9); sub-2 N differences (e.g., S1 vs. S3 $\Delta F_{\min}$=1.2 N, CI_{95}(S1)≈±0.7 N, CI_{95}(S3)≈±1.6 N) remain non-significant. Produced by `scripts/replicate_ablation_n10.py`. Raw outputs: rows L2 and S0–S5 in `outputs/paper_statistics/ablation_n10_results.json`; row L2.5 in `outputs/paper_statistics/ablation_n10_L2_5.json`; rows L0 and L1 in `outputs/paper_statistics/push_ci_all_ablations.json`. All at N=10 under the same protocol.

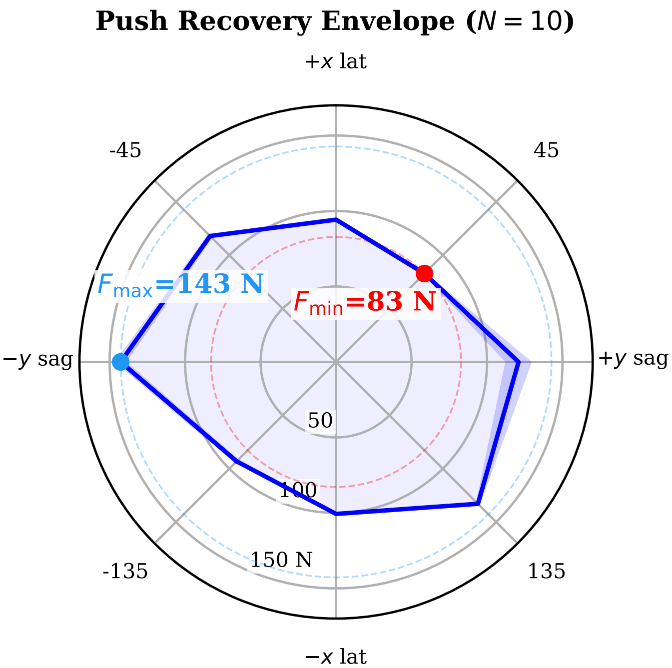

Fig. 4. Omnidirectional push envelope of ACC. This is the *same* run as row S1 of Table IV, not a separate measurement: 8 bearings, N=10 repetitions each, threshold located by binary search over 10 – 160 N to a 5 N tolerance, so the plotted minimum is by construction the F_{min} of that row. Solid curve: per-bearing mean survivable push magnitude; shaded band: ±1 standard deviation over the ten repetitions. The radial axis is linear in force and originates at 0 N, so radial distance is proportional to margin. The envelope is strongly anisotropic, spanning 82.8 – 142.7 N—a 1.7× ratio between the weakest and strongest bearing. The ordering is −90° (142.7 N) > 135° (132.9 N) > 90° (120.9 N) > −45° (118.0 N) > 180° (100.7 N) > 0° (94.3 N) > −135° (93.1 N) > 45° (82.8 N). Two regularities are visible. First, the actuated axis is the strong one: bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so ±90° are purely sagittal and 0°/180° purely lateral (Sec. V-A). The two sagittal bearings average 131.8 N against 97.5 N for the two lateral ones, a 1.35× advantage. This is the expected asymmetry for a wheeled biped: sagittally the wheels accelerate the contact patch back under the CoM and convert the impulse into a controlled excursion, whereas laterally no actuator produces ground force at all and the only restoring authority is the static $w_{track} = 0.278$ m track width plus hip-roll PD. We compare axis *means* rather than ranks because the raw ordering is contaminated by the second regularity. Second, the envelope is measurably *chiral*: 135° survives 40 N more than −135°, and −90° 22 N more than 90°. An independent earlier sweep of a neighbouring configuration reproduces both asymmetries with the same sign and comparable magnitude, so this is a systematic property of the plant-plus-controller rather than sampling noise; we have not isolated its origin, and it is reported here as an observation rather than an explained effect (Sec. VI). F_{min} (red) is the bearing-wise minimum and is the figure of merit quoted throughout the paper, because it is the magnitude survived in *every* direction; F_{max} (blue) is shown only to indicate the spread.

causes 10/10 falls at pure idle. S3 (asymmetric EMA) and S2 are each individually required for ringdown.

C. Flight Recovery and Terrain Adaptation

ACC's two-channel assembly supports independent extensions without modifying the core anchor. Both extensions use the same 400 Nm/s rate limit and 100 Hz control rate.

Contact-loss recovery. ACC recovers to a settled stand from a 100 cm free drop (10/10, 24.0±1.2° peak pitch after touchdown) and from a 50 cm ledge drive-off (10/10, 27.1±2.0° peak landing pitch); the 60 cm drop and 20 cm ledge are

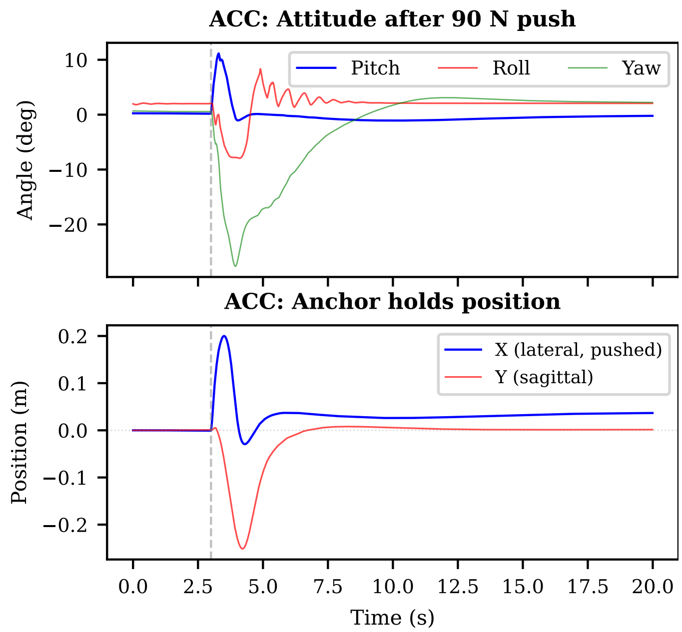

Fig. 5. Post-push recovery (90 N lateral, world $+x$, at t=3 s). **Top:** attitude angles; stabilization within 2 s. **Bottom:** planar CoM position, which separates the two axes cleanly. The lateral push couples through roll into a large *sagittal* excursion (−251 mm peak); the anchor drives that axis back to 1.37 mm of home and holds it there at 0.018 mm SD over the final 2 s—a 99.5 % recovery. On the pushed lateral axis, which has no actuator authority (Sec. V-A), the 200 mm peak excursion decays only to a permanent 36.2 mm offset, 18 % of the peak. The contrast between the two traces is the clearest single illustration of what the anchor does and where it does not act.

correspondingly milder (Table V). The two flight ablations, however, give a more nuanced picture than the mechanism's design rationale suggests.

Removing the flight *attitude* PD (the reaction-flywheel term $2.0\theta + 0.5\dot{\theta}$, spin-down damping retained) is a real but modest loss: peak pitch rises from 24.0 to 27.8° on the 100 cm drop with one fall in ten, and from 27.1 to 31.6° on the 50 cm ledge with no falls. Removing the airborne wheel-torque override *entirely*—so the ordinary wheel–inverted-pendulum (WIP) loop keeps driving the wheels through flight—does not degrade either condition at all: it *lowers* peak pitch to 13.0±0.5° (drop) and 10.2±0.4° (ledge) at 10/10 survival, because free wheels driven by the pitch-feedback loop are themselves an effective reaction flywheel. The override's benefit appears only at the upper end of the envelope: at 150 cm the two configurations survive equally (2/5 settled either way) but the override holds peak pitch to 34.0±2.1° against 53.6±35.0° without it—a 16× reduction in spread, with one uncontrolled trial exceeding 90°. At 200 cm both configurations fail. The flight override therefore **bounds landing-attitude variance near the envelope limit** rather than enabling the recovery; at the 50 – 100 cm scales reported here ACC's balance core alone suffices. The distinction is worth drawing because the mechanism was introduced to prevent a nose-down crash observed in an earlier controller revision, a failure mode no longer reachable in the present design.

Per-leg terrain adaptation. Here the ablation is unambiguous. With per-leg adaptation, straddle roll on the one-wheel curb is 2.5±0.04° (10 cm), 3.5±3.2° (15 cm) and 14.9±0.1° (20 cm), all 10/10. Replacing `split()` with a symmetric posture (both legs commanded to the same height—the pre-adapter behaviour) causes a fall at *every* tested curb height, 0/10 in all three cases: the torso rolls past 60° within 0.8 s of curb entry and the robot is on the ground by 2.3 s. The height split is thus load-bearing for this task in a way the flight override is not. The closed-loop leveling trim that rides on top of the split is a second-order refinement of the same mechanism: disabling it alone leaves the robot upright on the 10 cm curb (3.2±0.08°, 10/10) but degrades the 15 cm curb to 11.1±1.6° roll with 6/10 settled—the trim corrects the residual error of the non-linear height→leg-length table, which grows with step height. At 20 cm the limit is one of joint authority, not of the split geometry. Evaluating (21) at h^{cmd}=0.404 m gives commanded leg lengths of 0.350 m (flat) and 0.150 m (curb), i.e. a difference equal to the full 20 cm step with *zero* kinematic residual—the posture pair can span the curb exactly. What tightens is the joint budget: the compressed leg's commanded knee angle grows 2.18 rad (10 cm) → 2.39 rad (15 cm) → 2.58 rad (20 cm) against a hard limit of 2.70 rad, leaving only 0.12 rad (6.6°) of headroom, and the commanded CoM height of that leg, 0.265 m, sits within 1.1 cm of the extrapolation floor $z_{\mathrm{lo}}-10$ cm=0.254 m. Because the commanded posture is exact, the measured 14.9° straddle roll is entirely *tracking* error: in this deeply flexed configuration the leg carries its load at the shortest gravitational moment arm the PID gains were never re-tuned for, and it settles below its target. Adaptation therefore still prevents leg-fold (10/10 survival), but the torso is left tilted; the tight ±0.1° std reflects a deterministic, saturation-dominated equilibrium rather than genuinely low run-to-run variability.

The gate structure isolates each extension to its physical regime without re-tuning the anchor, and both extensions cost three lines of controller code each. Their measured *necessity*, however, differs sharply: the per-leg height split is required for the curb task at every tested height, whereas the flight override is a variance-bounding refinement that only pays at the top of the drop envelope.

TABLE V

CONTACT-LOSS RECOVERY AND PER-LEG TERRAIN ADAPTATION, WITH THE CORRESPONDING MECHANISM ABLATIONS. ALL AT 400 Nm/s RATE LIMIT, 100 Hz CONTROL. SURVIVAL IS THE *strict settle* VERDICT (FLAT PITCH/ROLL WINDOW, TAIL HEIGHT ERROR ≤12 mm, RESIDUAL SPEED ≤0.05 m/s), NOT MERELY STAYING UPRIGHT.

Contact-Loss Recovery			
Scenario / ablation	N	Peak Pitch (°)	Settled
Drop 100 cm	10	24.0±1.2	10/10
Drop 60 cm	10	18.2±1.0	10/10
Ledge 50 cm	10	27.1±2.0	10/10
Ledge 20 cm	10	19.6±0.3	10/10
−Attitude PD, drop 100 cm[a]	10	27.8±0.3	9/10
−Attitude PD, ledge 50 cm[a]	10	31.6±2.2	10/10
−Flight override, drop 100 cm[b]	10	13.0±0.5	10/10
−Flight override, ledge 50 cm[b]	10	10.2±0.4	10/10
Drop 150 cm (envelope edge)	5	34.0±2.1	2/5
−Flight override, drop 150 cm[b]	5	53.6±35.0	2/5
Per-Leg Terrain (One-Wheel Curb, 0.15 m/s)			
Configuration	N	Straddle Roll (°)	Settled
Full adaptation, 10 cm	10	2.5±0.04	10/10
Full adaptation, 15 cm	10	3.5±3.2	10/10
Full adaptation, 20 cm	10	14.9±0.1	10/10
−Leveling trim, 10 cm[c]	10	3.2±0.08	10/10
−Leveling trim, 15 cm[c]	10	11.1±1.6	6/10
−Height split, 10 cm[d]	10	fall	0/10
−Height split, 15 cm[d]	10	fall	0/10
−Height split, 20 cm[d]	10	fall	0/10

±std over N trials with randomized ICs (±0.003 rad joint, ±1.5 mm CoM). Drop metric: peak $|\theta|$ from touchdown onward; ledge metric: peak $|\theta|$ through landing; curb metric: max $|\phi|$ while straddling the curb top. [a]Reaction-flywheel attitude PD zeroed; airborne spin-down damping retained. [b]Entire airborne wheel-torque override bypassed—the ordinary WIP loop drives the wheels through flight. [c]Closed-loop leveling integrator disabled; per-leg height split retained. [d]`split()` forced to a symmetric posture (both legs at the same commanded height), i.e. pre-adapter behaviour; roll diverges past 60° within 0.8 s of curb entry, so no straddle-roll statistic exists. Produced by `scripts/collect_flight_terrain_n10.py`; raw output `outputs/flight_terrain_n10/results.json`.

D. Comparison with Classical Baselines and a Preliminary RL Reference

To establish that ACC is not merely a tuned linear controller, we compare six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—together with a preliminary model-free PPO run reported as a reference point rather than as a seventh baseline (all share identical environment setups):

1) **LQR**—4-state TWIP [46], [47] with decoupled roll/yaw PD through a PID servo layer (~0.25 s lag).
2) **LQR + Integral AW**—Same as (1) plus pitch-error integral with conditional AW (clamped ±3.0 rad/s).
3) **LQR (Direct Torque)**—Re-optimized for direct-torque plant (τ_s=0, $k_{p,\mathrm{roll}}$=55.0), outputting torques directly.
4) **Coupled 6-State 3D LQR**—Extends (1) with roll states $(\phi, \dot{\phi})$.
5) **PI + AW (Direct Torque)**—DT-LQR + integral AW on fwd. position (conditional, ±15 cm gate); same leg/roll/yaw as DT-LQR.
6) **PPO reference (preliminary)**—Model-free policy trained from scratch for 5.4M environment steps in `BalanceEnv`: single seed, no domain randomization, no hyperparameter search. Included as a difficulty probe, not as a tuned RL baseline (Limitation 10).
7) **Coupled 6-State LQR (Direct Torque)**—Same 6-state model as (4) but outputs direct torques (bypassing PID servo, ~0.25 s lag removed). Gains re-derived for the direct-torque plant (K_{pitch}=113, K_{roll}=58 Nm/rad); leg posture via PD position control.

Direct-torque LQR removes the servo path, eliminating both the PID gains and the measured actuation lag. PI+AW adds integral anti-windup on forward position to DT-LQR—the

One-wheel curb (20 cm) — per-leg terrain adaptation

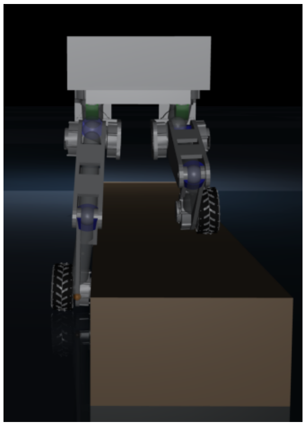

Fig. 6. One-wheel curb (20 cm). With the per-leg height split: 14.9±0.1° straddle roll, 10/10 settled. With a symmetric posture (split removed): the curb-side leg cannot fold far enough, roll diverges past 60° within 0.8 s of curb entry, 0/10.

standard engineering fix for a DC position bias, with windup protection. The anti-windup logic is verified independently by `scripts/validate_lqr_aw.py`. The PPO run serves a different purpose from the six classical controllers. It is included not to benchmark the state of the art in reinforcement learning, but to characterize how hard unaided model-free search is on the coupled pitch–height state space of a 10-DOF wheeled biped, and thereby to make explicit the value of the structured gate design ACC supplies in its place. Every gain used for the six classical baselines is listed in Table VI; we give them in full because the central claim of this subsection is a *negative* one—that no member of this family stands—and a negative result is only as strong as the tuning behind it.

Table VII reports the comparison. All six classical baselines—four derived from the 4-state TWIP model (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo, direct-torque)—fall within 0.60 s. Removing the PID servo path (4-state DT-LQR: τ_s=0, $k_{p,\mathrm{roll}}$=55.0, no integral) cuts torque by 2.5× (21.99 Nm → 8.76 Nm) and nearly doubles survival (0.32 s → 0.60 s), but *worsens* roll (ϕ_{RMS}=27.5° vs. 24.2°): the recovered authority is spent in the sagittal plane the model describes, not in the lateral one that ends the episode. Adding a forward-position integral (PI+AW: $k_{i,\mathrm{pos}}$=8.0, conditional ±15 cm gate) gives back 0.08 s of that gain rather than adding to it—the integral cannot accumulate usefully before the roll-induced fall—confirming the 4-state TWIP model is structurally insufficient for this articulated-leg platform. The preliminary PPO reference (5.4M steps, single seed) sustains single-height standing longer (3.20 s) but exhibits an 85 % fall rate across multi-height command transitions (54.2 mm height error). Read as a difficulty probe rather than a comparison, this is the observation the structured prior is meant to address: unaided search spends its budget rediscovering the pitch–height coupling that the posture map and the gate schedule encode directly. ACC achieves 0.060±0.006° pitch RMS, 0.010±0.001° roll RMS, 0.18±0.02 mm CoM Z, and 3.28±0.00 Nm torque

TABLE VI

BASELINE GAIN DERIVATION. ALL WEIGHTS FOLLOW BRYSON'S RULE; SEE THE NOTE BELOW THE TABLE.

Controller	Gains and derivation
LQR (4-state)	Q= diag(10, 2, 3, 0.3), R=0.8; roll and yaw closed by decoupled PD through the PID servo layer
LQR + AW	as above, plus a pitch-error integral with conditional anti-windup, clamped to ±3.0 rad/s
LQR, DT	4-state gains re-optimized for the zero-lag plant: τ_s=0, $k_{p,\mathrm{roll}}$=55.0
6-state LQR	Q= diag(10, 2, 3, 0.5, 3, 0.3), R= diag(0.8, 1.0), derived in continuous time and shared by the servo and DT variants
6-state LQR, DT	state feedback re-solved for the zero-lag plant: K_{pitch}=113, K_{roll}=58 Nm/rad (see Limitation 7)
PI + AW, DT	DT-LQR gains plus an integral on forward position: $k_{i,\mathrm{pos}}$=8.0 rad/s/(m·s), clamp ±5.0 rad/s, conditional gate ±15 cm
PID servo layer	k_p=[55, 40, 70, 70, 4], k_d=[3, 2, 4, 4, 0], k_i=[0.8, 0.4, 1, 1, 0.1], ordered (hip roll, hip yaw, hip pitch, knee, wheel)
Servo lag	0.25 s, measured as the wheel-velocity 90 % rise time (≈ 25 steps at 100 Hz)

Bryson's rule throughout (Q_{ii}=$1/x_{i,\max}^2$, R=$1/u_{\max}^2$); the 46-configuration retuning sweep described in the text confirmed that no retuning within these ranges changes the outcome. The 4-state model is (pitch, pitch-rate, forward velocity, forward position); the 6-state model adds (roll, roll-rate). "DT" denotes direct torque, i.e. the PID servo path removed and τ_s=0.

RMS (N=10). A **46-configuration** sweep (dimensions as given in Table VI; script `scripts/sweep_lqr_roll_gains.py`) confirmed all tested LQR variants fail (100 %); a 6-state coupled LQR with direct torque actuation, removing the PID servo path, also fails—indicating platform complexity, not servo lag alone, is the dominant factor (see Limitation 7). Neither servo-lag removal nor preliminary PPO training (5.4M steps, single seed) resolved the height-adaptive posture trade-off in our experiments; whether a fully converged PPO policy closes this gap is not tested here.

1) Both control rates give the same outcome: Table VII reports every classical baseline at ACC's 100 Hz, so the headline comparison carries no control-bandwidth differential. Those gains were tuned for a 50 Hz loop, however, and a reader is entitled to ask whether running them off their design rate is what produces the gap. Table VIII answers that by giving both rates side by side.

Doubling the baselines' control rate does not rescue any of them. Fall rate remains 100 % in every cell, and survival time does not improve for a single variant—it *shortens* for five of six (LQR: 0.52 s→0.32 s; LQR-DT: 0.72 s→0.60 s; PI+AW: 0.72 s→0.52 s) and is unchanged for the sixth. The diagnostic quantity is roll: ϕ_{RMS} is 24.21° at both rates for the three servo variants, and rises from 21.22° to 27.48° for LQR-DT. Faster sampling improves the quantities the controller actually regulates—pitch RMS falls by roughly 6× for the servo variants (0.037°→0.006°) and τ_{RMS} drops from 92.1 Nm to 22.0 Nm—while leaving the lateral divergence that terminates the episode untouched. This is the signature of a

TABLE VII

CLASSICAL BASELINES AND THE PRELIMINARY PPO REFERENCE VS. ACC, CLEAN SENSORS. ALL ROWS AT 100 Hz CONTROL EXCEPT THE PPO REFERENCE, WHICH IS REPORTED AT THE 50 Hz RATE ITS POLICY WAS TRAINED AT. PARENTHESISED N IS THE EPISODE COUNT. SEE THE NOTE BELOW THE TABLE.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP (N=20)	0.32	0.006	24.2	205.8	21.99
LQR + AW (N=20)	0.32	0.006	24.2	205.8	21.99
LQR, DT (re-opt.) (N=20)	0.60	0.033	27.5	119.2	8.76
6-St. LQR (N=20)	0.32	0.006	24.2	205.8	21.99
PI + AW, DT (N=20)	0.52	0.181	26.1	171.3	10.42
6-St. LQR, DT (N=20)	0.16	0.001	21.2	128.5	14.44
PPO ref. (50 Hz) (N=20)	3.20	4.82	3.15	54.2	18.40
ACC (N=10)	**20.00**	**0.060**	**0.010**	**0.18**	**3.28**

LQR / LQR+AW / 6-St. use the PID servo; LQR (DT), PI+AW (DT) and 6-St. (DT) command torque directly; PPO is a 5.4M-step model-free run (preliminary, single seed, no domain randomization), reported as a reference point rather than a tuned RL baseline. The identical LQR / LQR+AW / 6-St. rows arise because all three command through the same PID servo path, whose response dominates before the gain differential between them can act; the coincidence holds at both control rates (Table VIII).
Between-episode standard deviations (omitted above for legibility): survival time is deterministic on all six classical rows (±0.00 s at N=20, seeds $\{0, 42, 123\}$); PPO: ±0.85, ±0.91, ±0.45, ±8.3, ±2.1; ACC: ±0.00, ±0.006, ±0.001, ±0.02, ±0.00. Per-metric between-episode dispersions for the classical rows are not retained in the rate-matched output (`outputs/rate_matched_baselines/results.json`), which stores episode-aggregated values.

All LQR and PI+AW variants fall within 0.60 s (100 %); the 6-St. DT variant falls deterministically at 0.16 s (zero trial-to-trial variance), confirming that removing the servo path with gains derived from a simplified model does not rescue stability. PI+AW (DT-LQR + integral AW on fwd. position) does not improve on DT-LQR at this rate—it is 0.08 s worse—because the integral cannot accumulate usefully before the roll-induced fall. The PPO reference achieves partial standing but suffers 85 % falls over multi-height transitions (H_{RMS}=54.2 mm). N=20 throughout; classical rows from `outputs/rate_matched_baselines/results.json`.

TABLE VIII

EVERY CLASSICAL CONTROLLER AT BOTH CONTROL RATES: ITS OWN 50 Hz DESIGN RATE AND ACC'S 100 Hz (THE RATE USED IN TABLE VII). PHYSICS IS 500 Hz THROUGHOUT; ONLY THE SUBSTEPS PER CONTROL STEP CHANGE. SEE THE NOTE BELOW THE TABLE.

	Surv. (s)		ϕ_{RMS} (°)		H_{RMS} (mm)	
Configuration	50 Hz	100 Hz	50 Hz	100 Hz	50 Hz	100 Hz
LQR, TWIP	0.52	0.32	24.21	24.21	370.6	205.8
LQR + AW	0.52	0.32	24.19	24.21	370.7	205.8
LQR, DT	0.72	0.60	21.22	27.48	135.0	119.2
6-St. LQR	0.52	0.32	24.20	24.21	370.6	205.8
PI + AW, DT	0.72	0.52	21.22	26.14	135.0	171.3
6-St. LQR, DT	0.16	0.16	21.05	21.19	117.2	128.5
ACC	—	**20.00**	—	**0.010**	—	**0.18**

Protocol. N=20 episodes per cell, seeds $\{0, 42, 123\}$, nominal scenario, clean sensors, 0 ms actuator delay. Episode length is held fixed in *seconds*, not steps: 1000 steps at 50 Hz and 2000 steps at 100 Hz both cover 20 s. The LQR feedback gains are continuous-time and therefore rate-independent; the rate change affects only the substep count, the drift and integral accumulators, the torque rate limiter, and (for the FD-linearized variant) the linearization step.
Outcome. Fall rate is 100 % in all twelve baseline cells. The 100 Hz columns are the ones carried into Table VII; the 50 Hz columns come from the same script and the same episodes, so the two rates differ only by the intended change. ACC is shown at 100 Hz only; the 50 Hz ACC run is analysed as a negative result in Limitation 5 and is not a usable row.
Produced by `scripts/collect_rate_matched_baselines.py` (`–control-hz` flag added to `scripts/eval_balance.py`); raw output `outputs/rate_matched_baselines/results.json`.

model-structure deficit rather than a bandwidth deficit: the 4-state TWIP model carries no lateral state at all, and the 6-state coupled variant carries roll but linearizes it about a single operating point, so no sampling rate supplies the missing dynamics. Reporting the baselines at 100 Hz in Table VII is the rate-matched choice, and it is deliberately not the flattering one for them: five of the six survive *longer* at 50 Hz. Both rates are given here so that the comparison can be read either way, and the fall rate is 100 % in all twelve cells.

The converse experiment—running ACC at 50 Hz—is a negative result: the controller does not survive the rate change, and it is recorded as a limitation (§VI, Limitation 5) rather than as a rate-independence claim. What Table VIII establishes is the direction that matters for the comparison: the classical baselines fail both at the rate for which they were designed and at ACC's rate.

E. Whole-Body Control Baselines

To address whether a tuned whole-body controller (WBC) could match ACC, we evaluated two WBC configurations that were developed and subsequently discarded during controller development: **(W1)** a task-priority quadratic-program (QP) WBC—minimizing COM height error, torso orientation error, posture damping, and wheel regularization subject to full-body dynamics and contact constraints—used as the *primary* balance controller; and **(W2)** the same WBC blended as an *assist* on top of ACC via $\boldsymbol{\tau}_{\mathrm{assist}} = \boldsymbol{\tau}_{\mathrm{ACC}} + \alpha \cdot (\boldsymbol{\tau}_{\mathrm{wbc}} - \boldsymbol{\tau}_{\mathrm{ACC}})$, evaluated both with the per-joint authority gate active and with it disabled. Both were evaluated under identical cloned simulation conditions (three-arm counterfactual: ACC baseline, WBC-only, ACC+WBC assist; 225 scenarios spanning fixed-height balance, height transitions, and push recovery at 20 – 120 N; 481 000 control steps; ACC profile `K2_JAX_DEDICATED_DEFAULT_V3`; no gain retuning). Every number reported below was measured *after* the four audited implementation defects (F1–F4, Appendix B) were repaired; no pre-repair measurement is used as evidence anywhere in this paper.

W1 (WBC as primary controller). The standalone QP-WBC falls in **225 of 225** scenarios; ACC falls in **none**. It holds the robot upright for 0.570 s (SD 0.001 s) and spends 97.3 % of every episode in a fallen state (468 179 of 481 000 control steps). The failure is *lateral*: roll RMS is 93.45° against ACC's 3.71° (25.2×) and peaks at 103.9°—past horizontal—while pitch RMS is only 2.33° against 0.52° (4.5×). Base height collapses from 0.540 m to 0.091 m. The fallen fraction is uniform across all five scenario suites (97.2 – 97.4 %), so the collapse is not specific to pushes or to commanded height changes.

The failure is invariant to the implementation defects. Repairing F1–F4 raised the QP success rate from 85.85 % to

99.99 % (480 950 of 481 000 steps) and brought commanded torque into a physically sane range (RMS 0.96 Nm, max 4.33 Nm), yet time-to-fall moved only from 0.52 s to 0.570 s. A correctly-solving WBC falls at essentially the same instant as a defective one. Nor is the failure an artifact of one task weighting: re-running the WBC-only arm at four task-weight modes over an identical scenario set (Table XII) moves the fallen fraction by 0.1 percentage points and upright time by 0.03 s. The best mode, `torso_priority`, improves pitch RMS by 40 % but *worsens* roll—within a single SO(3) orientation task carrying equal roll and pitch gains, the QP trades one axis against the other rather than fixing both.

Mechanism. The WBC's posture task is a velocity damper, not a position tracker. Its reference $\mathbf{q}_{\text{ref}}$ is rebuilt from the currently measured joint vector at every control step, so $\mathbf{q} - \mathbf{q}_{\text{ref}} \equiv \mathbf{0}$ identically and the task collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ with $k_{d,\text{posture}}$=2.0. It can slow a postural drift but cannot reverse one. Combined with the absence of gravity-compensation feedforward, nothing in the task stack restores the legs once a lateral excursion begins—which is why the collapse is lateral, why it is insensitive to the task weighting, and why repairing the solver does not delay it. Critically, this is not solely a tuning or implementation defect: the WBC lacks the *regime-gating* that ACC's proximity-gated anchor provides. A WBC distributes torque optimally for a fixed task set—it cannot express qualitatively distinct physical regimes (quiet stance, push ringdown, large excursion) as distinct control modes. The precision–recovery trade-off ACC resolves through spatial gating and asymmetric temporal envelope-following is not expressible as a task-priority QP objective; even a bug-free WBC would lack the structured prior that gates the anchor integral, and would therefore face the same global-integral windup that our ablation S4 (no proximity gate, Table IV) confirms is fatal for sub-millimeter precision.

W2 (WBC as assist on ACC). We ran the assist in both of its available configurations, which together constitute a gate ablation (Table IX).

Gate disabled (α=0.25 fixed, per-joint authority capped at 20 % of the ACC torque; this is the configuration behind the 225-scenario campaign). The QP solved on 100.0000 % of steps with zero fail-closed events, so WBC torque entered the blend unconditionally. The assist arm never falls—ACC carries it—but it degrades ACC on every metric except one: pitch RMS 1.295° vs. 0.520° (2.5×), yaw drift 12.56° vs. 6.69° (1.9×), planar drift 0.199 m vs. 0.099 m (2.0×), height RMS 0.518 m vs. 0.540 m, and torque RMS 3.67 Nm vs. 3.18 Nm.

Gate enabled (per-joint authority $\alpha_j = \alpha_{\max} \cdot g_{\text{stability}} \cdot g_{\text{height}} \cdot g_{\text{push}} \cdot g_{\text{divergence}} \cdot A_j \cdot K_{\text{role},j}$, $\alpha_{\max}$=0.25). We instrumented the gate to record α and every factor at every control step. On the one height variant where the gate opens at all, the admitted authority is $\bar{\alpha}$=0.0061 (max 0.0146): 2.5 % of the permitted $\alpha_{\max}$, peaking at 5.8 %. On the remaining four variants the height factor g_{height} falls below its 1×10^{-3} cutoff on the commanded height alone and α is identically zero (Appendix B, Table XIII). The gated assist is consequently indistinguishable from ACC alone on identical scenarios—pitch RMS 0.519° vs. 0.520°, torque RMS 3.154 Nm vs. 3.154 Nm, height RMS 0.5412 m vs. 0.5412 m—agreeing to four significant figures because the gate admits almost nothing.

The two configurations bracket the assist: where WBC authority is admitted it degrades ACC, and where it is harmless it is inactive. No setting of the gate makes the assist simultaneously active and beneficial.

The one metric WBC improves. Roll is the exception in both configurations: with the gate disabled, roll RMS improves from ACC's 3.71° to 3.20° over the full batch (−14 %), and from 3.55° to 2.90° on the matched subset (−18 %). The improvement is systematic rather than sampling noise: it appears in both the full batch and the matched subset. ACC regulates the lateral axis with decentralized hip-roll PD, whereas the WBC uses a centralized orientation task that exploits full mass-matrix coupling; on that axis alone, the centralized formulation is the better regulator, and blending the two writes both into the hip-roll joints from incompatible premises. What it cannot do is stand: its posture stack has no position authority, so buying 18 % of roll costs 2.5× the pitch error and 2.0× the drift. A centralized lateral task is therefore a credible direction for improving ACC's weakest axis—the 97.5 N mean lateral push bearing of Limitation 1—but not through torque-level blending with a posture stack that cannot hold a posture.

Timing. After the F1–F4 repairs the QP solve costs 0.169 ms mean and 0.30 ms p99 over 481 000 solves (OSQP, offline Python/NumPy path), comfortably inside the 10 ms control period. Compute is therefore not the constraint. The offline matrix-assembly path surrounding the solver is un-optimized Python, and production C++ whole-body controllers run at 400 – 1000 Hz on embedded hardware. The assist is excluded on control grounds alone: it degrades every stability metric but one wherever it is active, and does nothing wherever it is not.

Summary. Neither standalone WBC nor WBC-as-assist resolves the precision–recovery trade-off on this platform. The trade-off is resolved by regime-gating—the proximity-gated anchor and asymmetric envelope follower—which WBC's task-priority formulation does not express. ACC's conditional gates therefore provide the essential structured prior; the WBC gap is one of structural expressiveness, not torque optimality.

F. Architectural Analysis

We evaluate ACC under four dimensions: noise/delay robustness (Table X), gate parameter sensitivity, torque rate-limit invariance, and compound-disturbance recovery.

Robustness to sensor noise and actuator delay. A factorial sweep over four noise levels and three delays (N=5 per cell), extended along the clean-delay axis to 150 ms, reveals two distinct delay regimes: a **performance threshold** where $F_{\max}$ bifurcates (94 – 96 N at 0 ms vs. 41 – 50 N at any delay, idle RMS 5.5×), and a **stability threshold** above 150 ms—the robot survives all tested clean delays (10 – 150 ms, N=5, zero falls) with stable post-bifurcation performance ($F_{\max}$≈49 N,

TABLE IX

THREE-ARM COUNTERFACTUAL UNDER CLONED INITIAL CONDITIONS: ACC ALONE, STANDALONE QP-WBC, AND WBC BLENDED AS AN ASSIST ON ACC WITH THE AUTHORITY GATE DISABLED AND ENABLED. ALL ARMS POST-REPAIR (F1–F4).

Metric	ACC	WBC	ACC + WBC assist gate off	ACC + WBC assist gate on
Full batch: 225 scenarios, 481 000 *control steps*				
Scenarios with a fall	0/225	225/225	0/225	—
Episode fallen (%)	0.0	97.3	0.0	—
Upright time (s)	full	0.570	full	—
Pitch RMS (°)	0.52	2.33	1.295	—
Roll RMS (°)	3.71	93.45	**3.20**	—
Roll max (°)	8.37	103.89	7.64	—
Yaw drift (°)	6.69	6.80	12.56	—
Base height RMS (m)	0.540	0.126	0.518	—
Final height (m)	0.538	0.091	0.515	—
Planar drift (m)	0.099	0.165	0.199	—
Torque RMS (Nm)	3.18	0.96	3.67	—
Max abs. torque (Nm)	9.70	4.33	10.74	—
QP success (%)	—	99.99	100.00	—
Gate-open subset: 5 scenarios, 10 775 *steps*				
Pitch RMS (°)	0.520	—	1.284	0.519
Roll RMS (°)	3.545	—	**2.903**	3.540
Yaw drift (°)	5.479	—	6.533	5.463
Base height RMS (m)	0.5412	—	0.5190	0.5412
Planar drift (m)	0.0674	—	0.1884	0.0677
Torque RMS (Nm)	3.154	—	3.664	3.154

"Full" upright time denotes no fall in any episode (20 s for fixed-height and height-transition scenarios, 21.55 s for push scenarios). Base height RMS is the root-mean-square of *absolute* base-z, so a larger value means the robot stayed up: WBC's 0.126 m is a collapsed torso, not tighter tracking. Torque RMS is likewise not an efficiency measure here—WBC's 0.96 Nm is the torque of a machine lying on the ground. Roll is the sole metric on which the assist beats ACC (bold); see §V-E. The gate-open subset is the `low_small` height variant, the only one of five on which g_{height} does not collapse (Table XIII); the gated arm was not run on the other four because its gate is identically closed there.

TABLE X

ROBUSTNESS TO SENSOR NOISE AND ACTUATOR DELAY. IDLE RMS: LATERAL (WORLD x) COM STD. OVER 20 s AFTER 3 s SETTLE (EMA_0=0 TRANSIENT), N=5. $F_{\max}$: BINARY-SEARCH LATERAL (WORLD +x) PUSH, ±5 N TOLERANCE, N=5.

Condition	Idle RMS (mm) Mean	Idle RMS (mm) ±Std	$F_{\max}$ (N) Mean	$F_{\max}$ (N) ±Std
Clean, 0 ms delay	0.42	0.03	96	1
Clean, 10 ms delay	2.30	0.16	47	2
Clean, 30 ms delay	2.62	0.22	46	3
Clean, 50 ms delay	2.61	0.42	49	8
Clean, 100 ms delay	2.23	0.25	48	4
Clean, 150 ms delay	2.40	0.22	45	1
Low noise, 0 ms delay	6.33	2.57	96	1
Low noise, 10 ms delay	12.82	3.93	47	2
Low noise, 30 ms delay	7.31	2.51	50	1
Med noise, 0 ms delay	34.00	11.28	94	3
Med noise, 10 ms delay	41.66	23.24	45	10
Med noise, 30 ms delay	35.10	16.83	41	10
High noise, 0 ms delay	3.00	0.55	95	3
High noise, 10 ms delay	3.00	0.55	95	3
High noise, 30 ms delay	— (fell)		— (fell)	

Noise (1-σ): low = 0.01 rad/s (ang. vel.), 0.01 m/s^2 (grav.), 0.001 rad (joint pos.); med = 5× low; high = 10× low. Two distinct delay regimes: **performance** bifurcation at ~10 ms ($F_{\max}$: 96 N→ ~47 N); **stability** threshold >150 ms (zero falls at all tested clean delays, N=5; post-bifurcation idle ≈2.5 mm, $F_{\max}$≈45–49 N). The apparent medium→high noise improvement (34 mm→3 mm) occurs because high noise holds EMA above threshold, disengaging the anchor into P-only mode. Non-monotonic low-noise RMS (12.82 mm at 10 ms vs. 7.31 mm at 30 ms) reflects resonance: 10 ms phase-shifts the anchor into intermittent engage/release; at 30 ms EMA stays above threshold (medium noise follows the same pattern).
Produced by `scripts/collect_robustness_sweep.py` (noise×delay grid, raw output `outputs/paper_statistics/robustness_sweep.json`) and `scripts/collect_delay_stability_sweep.py` (clean-delay stability threshold, raw output `outputs/paper_statistics/delay_stability_sweep.json`).

idle ≈2.5 mm). ACC's gates depend on physical state (position, velocity envelope, orientation), not tight timing; jitter that crosses the 10 ms performance threshold degrades push margin temporarily but does not cause a fall. High noise causes fall regardless of delay; the asymmetric EMA cannot distinguish noise from motion, forcing anchor disengagement. Idle repeatability degrades from 0.42 mm (clean) to 34.00 mm under medium sensor noise—an 81-fold loss. The comparison that makes this concrete is against the un-anchored baseline on the same axis: L0 holds R_{lat} = 3.675 mm (Table III), so under medium sensor noise ACC is roughly an order of magnitude *worse* than having no anchor at all. The mechanism is the one described above—the envelope gate cannot separate sensor noise from real motion, so it releases the integral and then re-engages it repeatedly—and the practical consequence is that ACC as presented requires a filtered velocity estimate, not a raw one. Two caveats attach to the absolute values in this table. First, its idle column is measured on the *lateral* axis (world x), not the sagittal one of Table III, so the two are not directly comparable in level; the like-for-like clean reference here is ACC's lateral R_{lat} = 0.730 mm, and the 0.42 mm entry is lower still because of the shorter settling allowance (3 s here vs. 5 s there): with EMA_0=0 the envelope gate opens fully at the start of the trial, so a window that begins earlier captures more of the strongly-anchored transient and less of the steady state. Second, the noise and delay *ratios* are unaffected by either issue, since every cell in the table is measured identically; it is the degradation factors, not the absolute millimeters, that this table is used for. Re-running the sweep with sagittal resolution and EMA_0=1.0 initialization would remove both caveats and is deferred.

Gate parameter sensitivity and tuning. Finite-difference gradients reveal a stark hierarchy: the **release coefficient** α_r **dominates** (sensitivity 220.4, 20× the next, 130× pitch gates); pitch stability thresholds (≈0) are effectively insensitive safety interlocks. For porting to a new morphology: (1) identify dominant ringdown period T_r from post-push decay; (2) set τ_{release}≈3–5×T_r; (3) set quiet-stance band 2–3× idle velocity noise floor; (4) set pitch gates conservatively (±12°, ±15 °/s); (5) tune k_p (fixed k_p=50 sufficed). A fixed k_p and anchor integral gain K_i require minimal adjustment across platforms; the critical parameter is τ_{release}.

Gate failure modes. Stuck-at-1 (g_{anchor}≡1) causes windup → fall (cf. S4); stuck-at-0 yields P-only limit cycle (cf. L0); stuckhalf-open triggers parametric resonance (cf. S5, $F_{\min}$=9 N). All three map to measured ablation rows (Table IV), providing implicit FMEA coverage.

Torque rate-limit invariance. The 400 Nm/s slew limit used throughout is not what sets ACC's push ceiling. Sweeping it over an 8× span, with each trial drawing an independent initial-posture perturbation (the same $\mathcal{N}(0, 0.005\,\text{rad})$ joint / $\mathcal{N}(0, 0.001\,\text{m})$ root draw as the push protocol of §V-B, N=5 per setting), gives lateral $F_{\max}$ of 92.5±1.1 N (100 Nm/s) and 94.4±0.0 N at 200, 400 and 800 Nm/s, with idle lateral RMS of 0.695±0.046, 0.738±0.031, 0.732±0.025 and 0.729±0.023 mm respectively—a 2 % spread in push margin and a statistically indistinguishable spread in idle precision. The limiter is genuinely active where it could matter: instrumenting the commanded torque during a 90 N push shows $|\dot{\tau}|$ saturating exactly at each configured bound (100, 200, 400 Nm/s; at 800 Nm/s the wheel channel still reaches the bound while the leg channels peak at 712 Nm/s). During quiet stance, by contrast, the peak commanded slew measured across the four settings is 0.51 – 1.66 Nm/s—60 – 190 times below even the 100 Nm/s bound—so idle precision cannot be slew-limited at any tested setting. The ceiling is therefore architectural rather than actuator-limited.[1]

Compound-disturbance recovery. The push protocol of §V-B holds the height command fixed. Because the posture map is height-scheduled, a commanded transition and a push compete for the same leg-torque budget, so the two disturbances are evaluated together here. Scenario A applies the standard lateral push impulse at the midpoint of a 1 s commanded height ramp of size Δh about the 0.404 m nominal CoM-z, and locates $F_{\max}$ by the same binary search; Δh=0 is the matched static control, run under the identical protocol so that the two differ only in the height command. Every trial draws an independent initial posture from the §V-B distribution, N=10 throughout. Table XI reports the result.[2]

Three findings follow, and they change the earlier reading of this experiment. First, *inside the calibrated posture band a simultaneous height command is nearly free*: a +5 cm rise costs 2.0 % of push margin (92.9 N vs. 94.8 N) and a −5 cm squat is marginally *better* than static (96.2 N, +1.5 %), because squatting lowers the CoM and shortens the toppling moment arm while the push arrives. Both effects are statistically resolvable but operationally negligible, and they bracket the static value—there is no shared-budget penalty at this amplitude. Second, the degradation is confined to the *extrapolated* region and is strongly asymmetric: a +10 cm rise collapses $F_{\max}$ to 18.2 N (−81 %), whereas the mirror-image −10 cm squat costs only 15 %. The asymmetry identifies the mechanism. Rising drives the knee toward extension, where the leg Jacobian is worst conditioned and the same joint torque produces the least vertical force, and it does so while raising the CoM; squatting moves in the opposite direction on both counts. The failure is therefore a property of the posture map's extrapolated upper end (§III-A), not of the anchor, and the operational consequence is a commanded-range recommendation rather than an architectural limit: transitions within the calibrated band are safe under push, transitions above it are not. The large peak-pitch spread of the +10 cm row (±14.0°, one trial reaching 48.9°) is consistent with trials sitting on the edge of the capture basin rather than comfortably inside it. Third, Scenario B tests sequential rather than simultaneous disturbance: a 90 N lateral impulse followed 2 s later by a 60 N counter-directed impulse, before the first ringdown has fully decayed. ACC survives **10/10** (Clopper–Pearson 95%CI [0.69, 1.00]) with an episode peak pitch of 11.89 ± 0.51°, essentially all of it from the first impulse: after the reversal the pitch peaks at only 2.78 ± 0.11° and never re-enters the 5° band used as the settle criterion, in every trial. The reversal is thus absorbed within the residual motion of the first recovery, which is the behavior the asymmetric envelope is designed to produce—$\tau_{\text{release}}\approx$1.5 s keeps the damping boost still partly engaged when the second impulse arrives.

TABLE XI
COMPOUND DISTURBANCE: LATERAL (WORLD $+x$) PUSH DELIVERED AT THE MIDPOINT OF A 1 s COMMANDED HEIGHT RAMP Δh ABOUT THE 0.404 m NOMINAL COM-z. N=10 PER ROW, INDEPENDENT INITIAL-POSTURE DRAWS, BINARY SEARCH OVER [10, 130] N. Δh=0 IS THE MATCHED STATIC CONTROL. ROWS WITHIN ±5 cm STAY INSIDE THE CALIBRATED POSTURE BAND (0.354 – 0.454 m); THE ±10 cm ROWS ARE EXTRAPOLATED.

Δh	Band	$F_{\max}$ (N)	vs. static	Peak pitch (°)
+10 cm	extrap.	18.2 ± 5.2	−81 %	9.9 ± 14.0
+5 cm	calib.	92.9 ± 0.8	−2.0 %	15.9 ± 0.5
0 (static)	—	**94.8 ± 0.8**	—	14.0 ± 0.2
−5 cm	calib.	96.2 ± 0.0	+1.5 %	11.9 ± 0.5
−10 cm	extrap.	80.7 ± 6.0	−15 %	12.1 ± 2.3

Mean±SD. All differences from the static control are significant under Welch's t-test at the Bonferroni-adjusted α=0.007 of §VI (p<4 × 10^{-4} for the ±5 cm rows, p<4 × 10^{-5} for the extrapolated rows). Peak pitch is measured over the whole episode at each trial's own $F_{\max}$, so it is a shape statistic at the survival boundary, not a severity ranking across rows. Produced by `scripts/compound_disturbance_corrected.py`; raw output `outputs/compound_disturbance/results_corrected.json`.

VI. DISCUSSION AND CONCLUSION

ACC achieves all demonstrated behaviors without neural network training, is hypothesized to provide a *reduced* sim-to-real transfer gap vs. learned policies—pending hardware validation [48]. Each gate was added after tracing a measured

[1] An earlier version of this sweep reported invariance over 200 – 800 Nm/s from six bit-identical rows. Both features were artifacts: the sweep wrote the bound in Nm/s divided by the control step, overshooting by 100× so that the limiter never bound, and it accepted a per-trial seed that it never used, so the five "trials" at each setting were the same deterministic run. The numbers above come from the corrected sweep (`scripts/sweep_rate_limit_corrected.py`, raw output `outputs/rate_limit_sweep/results_corrected.json`) and supersede it; the slew instrumentation is `scripts/probe_torque_slew.py`.

[2] An earlier version of this experiment reported a single 84 % degradation, described as a "0.65 m→0.50 m squat." Two defects invalidated it. The script in fact ramped the command *upward* from the 0.404 m nominal to 0.50 m—a ~10 cm *rise* into the extrapolated region, not a squat—and the label reflected an earlier base-z convention (§III-A). It also accepted a per-trial index that it never used, so the five "trials" were one deterministic run repeated and the reported spread was structurally zero. The corrected experiment supersedes it. Its Δh=+10 cm row reproduces the collapse, confirming the effect was real; what was wrong was its sign, its magnitude, and the claim that it applies to height transitions in general.

failure mode to its root cause: (i) bistable envelope; (ii) global-I windup; (iii) parametric pumping; (iv) hard-leash oscillation; (v) self-referential stiffness. All thresholds frozen prior to the ablation sweep (Table IV).

Statistical power. Welch's t-test, Bonferroni-adj. α=0.007 (7 comparisons); L0→L1 and L2→L3 significant (p<0.001, Cohen's d≈5.3, 6.4). The primary idle claims are stated as like-for-like ratios in Sec. V-A (59.4× sagittal repeatability, 22.6× planar repeatability, 5.0× lateral repeatability, all L0 vs. ACC at N=10 under one protocol; the headline is the sagittal figure, which is the actuated axis); the S5 collapse is unambiguous (0/10 survival at idle, 10/10 at every push magnitude). Flight/terrain: N=10, Clopper–Pearson 95%CI [0.69, 1.00].

Limitations. (1) The push envelope is anisotropic and chiral. The weakest bearing is the oblique 45° (82.8 N, 29 % below F_{med}), and the two lateral bearings average 97.5 N against 131.8 N for the two sagittal ones—the wheels can drive the contact patch back under the CoM sagittally but produce no lateral ground force. The chirality (Fig. 4) is reproducible but unexplained. (2) Oblique 45° ledge descent at ≥20 cm exceeds lateral balance authority. (3) All results are simulation-only (MuJoCo); hardware validation remains future work. (4) ACC exhibits two distinct delay thresholds: a **performance** bifurcation at ∼10 ms (F_{max} drops ∼95 N→∼45 N, the 0.5 ms end-to-end budget margin is for this threshold); and a **stability** threshold >150 ms (robot survives all tested clean delays, zero falls, N=5; see Deployment Risks). A jitter spike exceeding the performance threshold degrades push recovery temporarily but does not cause a fall—ACC's gates depend on physical state, not tight timing. (5) **ACC is not rate-portable downward.** A 50 Hz ACC evaluation with recomputed discrete coefficients (α_a≈0.58, α_r≈0.013, anchor leak 0.010) *failed*—the robot collapsed to a mean CoM height of 0.124 m (nominal 0.404 m), and the sub-millimeter idle RMS and 10 N push threshold recorded by that run are measurements taken on a fallen robot. Rate-independence of ACC's architectural contribution is therefore not claimed, and the responsible mechanism remains undetermined between two candidates: excitation of high-frequency plant modes that the 100 Hz loop suppresses, or the phase lag introduced by discretizing the asymmetric envelope at Δt=20 ms, where α_a≈0.58 leaves the attack filter barely more than one sample from deadbeat. Distinguishing them requires a loop-gain and phase-margin analysis of the linearized closed loop at both rates, together with a re-tuning of the ACC gains for the 20 ms step; neither has been performed. The practical consequence is a deployment constraint rather than a comparison flaw: ACC as tuned requires a 100 Hz loop, and the 9.5 ms end-to-end budget estimated below leaves no room to relax it. (6) **The commanded height range is bounded by the posture map, not by the anchor.** Within the calibrated band (0.354 – 0.454 m CoM-z) a commanded transition costs at most 2 % of push margin, but a +10 cm rise into the extrapolated region collapses F_{max} by 81 % (Table XI); the mirror-image squat costs only 15 %. Operating above the calibrated band under disturbance is therefore unsafe, and extending the band requires additional posture calibration points rather than a controller change. ACC has not been evaluated over a wide commanded-height sweep in the manner of a dedicated height-tracking study; only the ±5 cm and ±10 cm steps of §V-F are measured. (7) Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %). A full-state LQR from finite-difference linearization of the complete 10-DOF MuJoCo plant was attempted but produced unreliable gain matrices due to ill-conditioned contact dynamics at the standing equilibrium ($\kappa(A)$≈10^{10}); robust linearization of articulated-leg contact dynamics at unstable equilibria is an open challenge. The hip-roll coupling $b_{r,h}$= − 5.0 in the 6-state LQR is empirical; a sensitivity sweep is deferred. (8) **A 27** mm **sagittal parking offset persists on the very axis the anchor regulates; it is a finite-DC-gain effect, and it is removable.** ACC's position integral acts on the sagittal axis, yet the robot settles −27.0 mm from the commanded home along that axis and then holds the offset to within 0.009 mm SD across ten trials (Table III). Sagittal *repeatability* is excellent (0.294 mm), so the deficit is a static bias rather than motion, and it is largely insensitive to the gate structure (every subtractive ablation that survives lands within 0.2 mm of it). Earlier drafts attributed it to a mismatch between the latched home and the true torque-balance equilibrium, compounded by the CoM-z versus base-z convention gap (§III-A). Direct instrumentation of the sagittal torque decomposition (Appendix C) shows that hypothesis to be wrong on the dominant term and right only on the remainder. At the parking point the gravity moment about the wheel axle is 0.0040 Nm (CoM lever 0.050 mm)—a settled wheeled inverted pendulum necessarily parks its mass over the contact line—and the anchor integral sits at 0.198 Nm, 10.0 % of its 2.0 Nm clamp, so the offset is neither gravity-driven nor a windup or saturation artifact. What actually balances is a constant pitch-loop demand of −1.259 Nm, produced because the feedforward pitch reference commands +3.476° where the robot in fact balances at +2.034°. The position loop must cancel that torque, and because the anchor integral is *leaky* its DC gain is finite ($k_{I,\text{dc}}$=$k_i\Delta t(1-\lambda)/\lambda$=7.96 Nm/m, one fifth of k_{pos}), so a nonzero error is required to produce it: Eq. (27) predicts 24.88 mm against 24.99 mm measured, and tracks a 40× gain sweep and a five-point height sweep to within 0.92 %. Only the residual 1 – 3 mm is a frame-convention gap. Two corrections were measured, neither yet promoted into the shipped profile because every table in this paper was measured on it: recalibrating the feedforward pitch reference removes 92.5 % of the offset (−27.0 → −2.0 mm) with F_{min} unchanged (82.5 N vs. 82.8 N, p=0.55) but a borderline 4.5 N reduction in F_{med} (p=0.051); raising k_i from 4 to 160 removes 80 % at no measurable push cost (p≥0.32). Absolute positioning accuracy as published is therefore limited to ∼27 mm, but the limit is a design-parameter choice with a closed-form price, not an unexplained residual. (9) Architectural generality across morphologies remains to be demonstrated. (10) Push ablation at N=10 has min. detectable effect ∼3.2 N; sub-

2 N differences non-significant (see Statistical power). The model-free PPO run (5.4M steps, single seed, no domain randomization, no hyperparameter search) does not meet the standard of a tuned reinforcement-learning baseline and is not presented as one; it is a difficulty probe on the unaided search problem, and the 85 % fall rate is reported as observed rather than as an upper bound on RL performance for this platform. A properly resourced learning comparison—multiple seeds, domain randomization [49], [50], a tuned reward—lies outside the scope of a structured-prior contribution and remains open; the natural form of that comparison is residual learning over the prior rather than learning against it. A task-priority QP whole-body controller was benchmarked both standalone and as an assist on ACC and did not resolve the precision–recovery trade-off (§V-E). One caveat remains open there—roll RMS is the single axis on which the WBC blend beats ACC (14 % over the full batch, 18 % on the matched subset), so the centralized formulation is not without merit on that axis; what the evidence establishes is that its advantage cannot be harvested by torque-level blending. A tuned convex MPC on the full 10-DOF plant remains deferred because finite-difference linearization at the standing equilibrium is ill-conditioned ($\kappa(A)\approx 10^{10}$, Limitation 7), so robust full-plant MPC is an open prerequisite rather than a missing comparison.

Deployment Risks and Mitigations. ACC uses raw torque, 400 Nm/s rate limit, standard sensors (IMU + encoders). We distinguish two latency thresholds: a **performance** threshold (~10 ms) where $F_{\max}$ bifurcates 96 N→~47 N, and a **stability** threshold (>150 ms, clean sensors, N=5, zero falls) where the robot actually falls. ACC's gates (proximity, envelope, pitch) depend on physical state, not timing precision—the architecture is intrinsically delay-tolerant for stability. The 150 ms+ stability threshold is consistent across all physical gating mechanisms: the proximity gate (spatial, 15 cm), asymmetric envelope (temporal, $\tau_r \approx 1.5$ s), and damping boost are all state-driven rather than schedule-driven.

Estimated non-RTOS worst-case budget: sensor readout ~1 ms + state estimation ~2 ms + ACC computation ~3 ms + actuation latency ~3.5 ms = ≈9.5 ms end-to-end. The 0.5 ms margin below the 10 ms performance threshold means any single jitter source (OS scheduling preemption, bus contention) can degrade push recovery margin temporarily; the stability margin remains >140 ms. On a real-time OS with a dedicated balance-loop microcontroller (e.g., 1 kHz RT Linux or bare-metal MCU), jitter is typically <0.1 ms, recovering 2 – 4 ms of budget and making the performance margin comfortable. Neither scenario has been validated on hardware—latency characterization with an oscilloscope and jitter histogram is required before deployment. The discarded WBC-assist variant (§V-E) is excluded from the hardware path on control grounds, not compute grounds: post-repair its QP solve costs 0.169 ms mean and 0.30 ms p99 over 481 000 solves and would fit the budget, but admitting its torque degrades pitch by 2.5× and planar drift by 2.0×, while gating it to the point of harmlessness also gates it to 2.5 % of its permitted authority. Hardware degradation: 0.294 mm sagittal repeatability → projected 1 – 3 mm (encoder quantization, backlash, stiction, tire compliance, IMU noise; cf. Table X). Staged commissioning: static torque → standing → push recovery → flight (tethered) → terrain.

We presented ACC, a two-channel torque assembly where a PD balance core is augmented by a proximity-gated anchor integral achieving 0.294±0.015 mm idle repeatability on the sagittal (actuated) axis—a 59× improvement over the proportional-only baseline under one unified protocol (N=10; Table III)—with omnidirectional push survival ($F_{\min}$=83 N, F_{med}=117 N). Absolute accuracy on that axis is an order of magnitude worse: the robot parks 27.0 mm from the commanded home, holds that offset to 0.009 mm, and closing it is the clearest remaining work on the design. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %); adding integral action to LQR-optimal feedback does not prevent the fall, confirming ACC's structured prior as the essential contribution. ACC tolerates moderate noise, is insensitive to the actuator slew bound over an 8× span (100 – 800 Nm/s: 2 % spread in $F_{\max}$, no resolvable change in lateral idle precision), exhibits a push-performance delay bifurcation at ~10 ms with stability maintained beyond 150 ms, and handles contact loss and terrain without neural network training. The contribution is thus a *structured classical prior*: an interpretable controller whose gating encodes the plant's measured failure modes, and whose intended downstream role is to supply the nominal base action of a residual-learning stack rather than to compete with a fully resourced learned policy. Future work: hardware validation, EMA settle-time characterization, a properly resourced learning comparison (multiple seeds, domain randomization, tuned reward), and extending conditional activation to locomotion and to residual RL over the prior.

Code and Data Availability

The controller, the MuJoCo model, every evaluation script referenced above, and the raw result files backing each table are available at https://github.com/thuong180702/Wheeled-bipedal-robot-simulation (branch `main`; a release tag will be minted at publication). Each main results table names its producing script and its raw result file in the note directly beneath it (Tables III, IV, V, X and XI); the remaining sweeps are cited inline at the point where they are discussed. The two superseded experiments of §V-F ship both the original and the corrected result files, so the correction itself is auditable. The raw per-step records behind the whole-body-control results of §V-E—the 225-scenario three-arm campaign, the task-weight ablation, the gated and ungated assist runs, and the per-step gate trace—are collected under `outputs/wbc_postfix_evidence/`, whose `README.md` maps each file to the table it produces. The torque decomposition, gain sweep, height sweep and push-validation records behind Appendix C are collected the same way under `outputs/parking_offset_evidence/`.

Appendix A
Why Saturation-Triggered Anti-Windup Is Insufficient

Saturation-triggered AW triggers on actuator saturation; on this platform, position error grows to 10 – 30 cm during a push while wheels stay within their 20 rad/s range, so saturation-triggered AW never activates. A static dead-zone either over-restricts or under-restricts; ACC's proximity gate with continuous smoothstep avoids both. State-gated conditional integration is practiced in process control; ACC couples spatial gating with asymmetric envelope-detected damping, providing temporal gating absent in static-threshold schemes. AW-augmented baselines (LQR+AW, PI+AW) are evaluated in Table VII; broader AW taxonomy comparison is deferred.

Appendix B
WBC Baseline Implementation Details

A. Evidence Provenance

Every W1 and W2 number in §V-E comes from one post-repair campaign (225 scenarios, 481 000 control steps) executed after all four audited defects were fixed; the full metric set is Table IX. The raw artifacts of the earlier pre-repair evaluation were not retained—only its summary report (`docs/validation/V3_vs_V3_Assist_comparison_report.md`) survives—so exactly two pre/post deltas are quoted, both recorded in that report: QP success rate (85.85 % → 99.99 %) and time-to-fall (0.52 s → 0.570 s). No other pre-repair quantity is quoted, compared, or relied on.

B. Scope of the Failure-Mechanism Claims

Only one failure mechanism is asserted for this baseline, the one verified directly in the source: the posture task is a velocity damper, not a position tracker (§V-E). Two further mechanisms attributed to the WBC in an earlier draft were checked against the implementation and withdrawn. The QP is *not* linearized at a fixed operating height—mass matrix, bias forces, CoM Jacobian and torso angular-velocity Jacobian are evaluated at the current state on every control step, and the module's only linearization is the standard, height-independent friction-cone pyramid. The wheels are *not* bare point contacts either: lateral no-slip and forward rolling constraints are implemented and were active (`rolling_mode=full_rolling_soft`) throughout the campaign. The architectural claim—absence of regime-gating in a fixed-task-priority formulation—rests on the invariance results of §V-E (defect repair, task-weight ablation), not on any deficiency specific to this implementation.

C. WBC Task-Weight Ablation

The standalone WBC's failure could in principle depend on the particular task weighting adopted. Table XII re-runs the WBC-only arm at four weightings over an identical scenario set. The spread between best and worst is 0.1 percentage points of fallen fraction and 0.03 s of upright time; all four collapse.

TABLE XII
WBC-only arm at four task weightings (`RANDOM_PUSH`, nominal height, seeds 201–205, identical scenarios). No weighting rescues the baseline.

Task weighting	Fallen (%)	Upright (s)	Roll RMS (°)	Pitch RMS (°)
Balanced (default)	97.4	0.570	92.86	2.51
Torso priority	**97.3**	0.590	95.28	**1.52**
Posture priority	97.4	0.570	93.52	2.37
CoM priority	97.4	0.560	93.15	2.42
ACC, same scenarios	**0.0**	full	**3.71**	**0.52**

Mode names are `balanced_default`, `torso_priority`, `posture_priority`, `com_priority` in `wheeled_biped/wbc/offline_task_stack.py`. The balanced default is w_{torso}=3, w_{com}=3, w_{posture}=1.5; each priority mode raises its named weight to 10. Torso priority gives the best pitch and the worst roll: torso attitude is a single SO(3) task carrying equal roll and pitch gains, so raising its weight redistributes error between the two axes instead of reducing both.

D. Assist Gate Decomposition

Table XIII shows why the gated assist is inert. The height factor $g_{\text{height}} = \exp[-((h-h_0)/\sigma_h)^2]$ uses h_0=0.53 m and σ_h=0.025 m, and is hard-zeroed below a 1×10^{-3} cutoff. Four of the five height variants in the scenario matrix command a base-z far enough from h_0 that the commanded-height term alone drives the gate shut before the measured height is even considered. On the one variant that remains open, the four state-dependent factors multiply to 0.461, and the per-joint agreement and role terms reduce the admitted authority by a further ∼19×, to $\bar{\alpha}$=0.0061.

TABLE XIII
Assist-gate decomposition. Top: the commanded-height term across the scenario matrix. Bottom: measured per-step gate factors on the one variant where the gate opens.

Height variant	Commanded base-z (m)	g_{height}
`high_small`	0.750	$2.3\times10^{-34}\to0$
`high_tiny`	0.670	$2.4\times10^{-14}\to0$
`nominal`	0.650	$9.9\times10^{-11}\to0$
`low_tiny`	0.630	$1.1\times10^{-7}\to0$
`low_small`	0.550	**0.53** (open)
Measured on `low_small`*,* 10 775 *control steps*		
g_{height}		0.527
$g_{\text{stability}}$		0.876
g_{push}		0.998
$g_{\text{divergence}}$		0.999
product of the four		0.461
$\bar{\alpha}$ admitted		**0.0061** (max 0.0146)
$\bar{\alpha}/\alpha_{\max}$		2.5 %

$\alpha_{\max}$=0.25. The remaining shortfall between the 0.461 factor product and the admitted 0.0061 is the per-joint agreement term A_j and role weight $K_{\text{role},j}$: the WBC's torque opposes ACC's on most joints, and disagreement attenuates the blend. h_0 and σ_h are hand-tuned constants of *ACC's* gate (σ_h was already widened from 0.015 m); they bound the authority ACC grants the WBC and are not a property of the WBC formulation, which is state-dependent and carries no fixed validity band.

E. Audited Implementation Defects (F1–F4) and Their Repair

A post-hoc audit of the WBC implementation found four defects (Table XIV). Reporting a baseline's failure while that baseline is known to be defective is not admissible evidence for an architectural claim, so all four were repaired and the repairs verified before the W1 numbers in §V-E were measured. Table XV lists the post-fix check applied to each defect and its measured outcome; the checks are executable (`scripts/verify_wbc_audit_fixes.py`) and their raw output is archived at `outputs/wbc_audit_fix_verification.json`. F4 is the one repair whose effect is partial; its residue is quantified below alongside the `PASS`.

TABLE XIV
AUDITED WBC IMPLEMENTATION DEFECTS (F1–F4). ALL FOUR ARE REPAIRED IN THE IMPLEMENTATION USED FOR THE RESULTS IN §V-E; SEE TABLE XV.

ID	Description	Effect on Metrics
F1	Contact Jacobian zero-gradient: leg joint columns = 0 (autodiff through cached FK).	Ground reaction forces produce zero moment on leg joints; $\boldsymbol{\tau}_{\text{wbc}}$ meaningless at hip-pitch/knee (K_{role}=0.60).
F2	Fast solver hardcodes `feasibility_only`—drops entire task stack.	QP finds minimum-norm $\ddot{\mathbf{q}}$; all task modes produce identical output.
F3	Offline eval passes base-z (0.536 m) as height cmd. instead of CoM-z (0.404 m)—13.2 cm error.	ACC gain schedule and feedforward at wrong operating point; drift/heading/centering gates disabled.
F4	Assist-gate frame conflict: g_{height} compared a CoM-z height reference against a base-z nominal h_0=0.53 m—a 13 cm frame error (§III-A).	$g_{\text{height}}{\approx}0$ at every test height; the assist gate is forced shut for a reason unrelated to the robot's state.

Source: `docs/validation/v3_wbc_assist_root_cause_audit.md`. Effects listed are those of the *unrepaired* code; each was removed by the corresponding fix in Table XV.

F. F4 Is Repaired but the Gate Still Collapses

The F4 repair removes the frame error: the gate now compares base-z against base-z. It does not, however, open the gate across the scenario matrix. With h_0=0.53 m and σ_h=0.025 m retained, four of the five height variants command a base-z that puts g_{height} below its 1×10^{-3} cutoff on the commanded term alone (Table XIII), and on the fifth the admitted authority is 2.5 % of $\alpha_{\max}$. The assist's equivalence to ACC is therefore not the robust bug-independent result an earlier draft took it for: it is a direct consequence of a gate that is shut or nearly shut everywhere on this matrix. The claim that survives is the one measured in both configurations of §V-E—running the assist *without* the gate admits WBC torque on 100 % of steps and degrades every stability metric but roll, which is evidence about the blend itself rather than about the gate's tuning.

APPENDIX C
STATIC TORQUE BALANCE OF THE SAGITTAL PARKING OFFSET

Limitation 8 reports a −27.0 mm sagittal parking offset on the axis the anchor integral regulates. This appendix replaces the hypothesis carried by earlier drafts with a measured torque decomposition, derives the offset in closed form, validates the derivation against a 40× gain sweep and a five-point height sweep, and reports the measured cost of two corrections. All runs use the idle protocol of Table III (identical seeds, 5 s settle, 20 s window) with the controller's internal sagittal torque decomposition logged alongside the gravity moment about the wheel axles (`scripts/diagnose_parking_offset.py`; raw records under `outputs/parking_offset_evidence/`). Sagittal is world y on this platform; the x columns of the raw records are lateral.

TABLE XV
POST-FIX VERIFICATION OF F1–F4

ID	Post-fix check	Measured result
F1	Analytic contact-point translational Jacobian compared element-wise against MuJoCo `mj_jac` at both wheel contact points; leg-column norms inspected for the zero-gradient symptom.	Max abs. error 4.2×10^{-8} (full), 3.4×10^{-8} (actuated block), vs. the audit's acceptance bound 1×10^{-3}. Ipsilateral leg columns nonzero (norms 0.042, 0.433, 0.403, 0.306 and 0.060); contralateral columns exactly zero, as the kinematics require.
F2	Task mode must reach the QP cost, i.e. the cached builder must be parameterized by `task_mode` rather than receiving the literal `feasibility_only`.	Builder takes `task_mode`; the single remaining literal call site is the task-independent constraint-only fallback, which carries no cost terms by construction.
F3	Batch runner must pass CoM-z, not base-z, as the ACC height reference.	Height reference is read from the scenario's `target_com_z`; the pre-fix base-z form is absent from the source.
F4	The adaptive-assist g_{height} gate must compare against its own base-z target rather than reusing the CoM-z reference.	Repaired: a separate base-z target is passed, so gate and controller agree on the frame. *Partial in effect*—see below.

Checks are executable: `scripts/verify_wbc_audit_fixes.py`; archived output `outputs/wbc_audit_fix_verification.json` (all four `PASS`). The frame distinction underlying F3/F4 is defined in §III-A.

A. The offset is neither gravity-driven nor a saturated integral

Two natural explanations fail on direct measurement (Fig. 7a, N=10, values averaged over the measurement window):

Gravity moment about the wheel axle	0.0040 Nm
CoM lever arm about the axle	0.050 mm
Anchor integral I	0.198 Nm (10.0 % of cap)
Integral clamp $I_{\max}$	2.0 Nm
Pitch-loop torque τ_{pitch}	−1.2587 Nm
Position-loop torque τ_{position}	1.2631 Nm
Adaptive bias trim τ_{trim}	0.0655 Nm
Controller position error e	−24.99 mm
Base parking offset $\bar{s}$	−26.96 mm

Gravity is not the load. The moment is 0.0040 Nm because the CoM sits 0.050 mm from the axle line. This is not a coincidence of tuning: a wheeled inverted pendulum that has stopped accelerating has necessarily brought its contact patch under its CoM, whatever its absolute position. The parking offset is a displacement of the whole machine relative to the commanded home, not a static lean, so no relation of the form $\tau_{\text{gravity}}(\bar{s}) \approx \tau_{\text{integral}}$ can hold.

The integral is not saturated. I reaches 10.0 % of its clamp during quiet stance and never approaches it, so the offset is not a windup or anti-windup artifact and cannot be relieved by raising the clamp. This is consistent with the ablation result of Table III, where removing the proximity gate entirely (S4) reproduces ACC bit-for-bit.

What does balance is τ_{pitch} against τ_{position}, to within 0.005 Nm.

B. Origin of the constant pitch torque

The sagittal pitch term is $\tau_{\text{pitch}} = k_{p,\theta}(\theta - \theta_{\text{ref}})$ with $k_{p,\theta} = 50\,\text{Nm/rad}$ at idle. Decomposing the commanded reference at the nominal standing height h=0.4040 m:

Feedforward equilibrium-pitch calibration	+3.4963°
Support-position outer loop (PD, k_p=0.8145 °/m)	−0.0203°
Low-band support term (gated off at this height)	0.0000°
Commanded reference θ_{ref}	+3.4760°
Measured body pitch θ	+2.0336°
Residual $\theta - \theta_{\text{ref}}$	−1.4424°
$\Rightarrow \tau_{\text{pitch}}$	−1.2587 Nm

The decomposition closes to four decimal places. The offset therefore originates in a 1.44° calibration error of the feedforward equilibrium-pitch map at this operating point, not in the anchor at all. Note that the outer loop, which does observe the 25 mm error, contributes 0.02° against the 1.44° it would need: its integral channel is disabled, so it too has finite DC gain.

C. Closed form

Writing the sagittal wheel torque during quiet stance as $\tau_{\text{pitch}} + \tau_{\text{position}}$, with the position law unsaturated ($|\tau_{\text{position}}| \approx 1.3\,\text{Nm}$ against a 4.0 Nm cap),

$$\tau_{\text{position}} = -k_{pos}\, e + I + \tau_{\text{trim}}. \tag{24}$$

The implemented anchor integral is *leaky*—Eq. (9) writes the ideal integral for clarity, but the discrete update applied at every step is

$$I_{k+1} = (1-\lambda)\big(I_k - k_i\,\Delta t\, g_{\text{prox}} g_{\text{env}}\, e_k\big), \tag{25}$$

with λ=5×10^{-3} per step (Table II). A leaky integrator is a first-order lag, not an integrator: at a fixed point with the gates open, Eq. (25) gives $I^* = -k_{I,\text{dc}}\, e$ with

$$k_{I,\text{dc}} = k_i\,\Delta t\,\frac{1-\lambda}{\lambda} = 4.0 \cdot 0.01 \cdot \frac{0.995}{0.005} = 7.96\,\text{Nm/m}, \tag{26}$$

which is one fifth of k_{pos}=40 Nm/m, not infinity. Substituting into Eq. (24) and imposing zero net sagittal torque:

$$e_{ss} = \frac{|\tau_{\text{pitch}}| - \tau_{\text{trim}}}{k_{pos} + k_i\,\Delta t\,(1-\lambda)/\lambda} = \frac{1.2587 - 0.0655}{40 + 7.96} = 24.88\,\text{mm}, \tag{27}$$

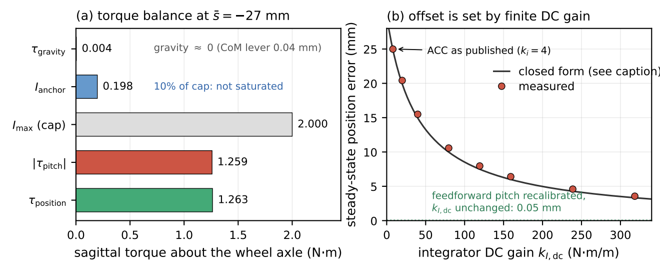

Fig. 7. Static torque balance of the sagittal parking offset. (a) Measured sagittal torques at the $\bar{s}$= − 27 mm parking point. The gravity moment about the wheel axle is 0.0040 Nm and the anchor integral sits at 10 % of its clamp, so the offset is neither gravity-driven nor a saturation artifact; the balance is between the pitch and position loops. (b) Steady-state position error against integrator DC gain $k_{I,\text{dc}}$ of Eq. (26), swept over $k_i \in [4, 160]$ at fixed leak. The curve is Eq. (27) evaluated with the baseline τ_{pitch}; using each run's own measured τ_{pitch} the closed form tracks all twelve gain and height points to within 0.92 %. The dotted line is the same controller with the feedforward pitch reference recalibrated and $k_{I,\text{dc}}$ left unchanged.

against 24.99 mm measured—an error of 0.45 %. The remaining 2.0 mm between e and the base offset $\bar{s}$ is the base-frame versus support-centre convention gap, which is the part of the earlier hypothesis that survives.

The conclusion is structural rather than incidental: *no element of ACC's sagittal position chain has infinite DC gain.* The anchor integral is leaky by design and the outer loop's integral channel is disabled, so a constant disturbance torque must be paid for with a constant position error. The 27 mm is the price of that design choice, in millimetres per newton-metre.

D. Validation across a forty-fold gain sweep

Sweeping k_i from 4 to 160 at fixed leak moves $k_{I,\text{dc}}$ over 40× and the offset follows Eq. (27) throughout (Fig. 7b):

k_i	$k_{I,\text{dc}}$ (Nm/m)	e meas. (mm)	e pred. (mm)	window jitter (mm)
4	**7.96**	**-24.99**	**-24.87**	**0.298**
10	19.90	-20.42	-20.34	0.302
20	39.80	-15.49	-15.54	0.147
40	79.60	-10.57	-10.67	0.208
60	119.40	-7.96	-8.01	0.168
80	159.20	-6.40	-6.41	0.094
120	238.80	-4.59	-4.58	0.114
160	318.40	-3.56	-3.56	0.066

Steadiness improves monotonically with k_i over this range rather than degrading, which is the opposite of the integral/steadiness trade-off measured at k_i=0 in Table III and indicates that the 3× steadiness cost reported there is paid at the first newton-metre of integral authority, not progressively.

E. The leak sets memory, not accuracy

Because $k_{I,\text{dc}}$ depends on both k_i and λ, reducing the leak also raises DC gain—but it is the wrong knob. Cutting λ from $5{\times}10^{-3}$ to $5{\times}10^{-4}$ raises $k_{I,\text{dc}}$ to 79.96 Nm/m and does reduce the offset to −16.31 mm, but window jitter degrades from 0.298 mm to 3.463 mm, a factor of 12, and the measured error (−14.34 mm) does not reach its DC prediction (−10.41 mm) because the forgetting time constant τ=$\Delta t/\lambda$ has grown to 20 s and no longer settles inside the measurement window. Since τ is independent of k_i, the two parameters separate cleanly: k_i sets DC gain, λ sets the memory horizon. They were conflated in the original tuning, where λ was chosen for a 2 s forgetting time and k_i was lowered from 8 to 4 to reduce limit cycling—which, per the sweep above, cost accuracy without buying steadiness.

F. Two measured corrections and their cost

The height dependence of the calibration error was checked on the five posture-calibrated height variants, applying the correction measured at each point. The nominal row is N=10 under the full idle protocol; the four off-nominal heights are single trials, whose run-to-run spread the nominal row bounds at 0.011 mm SD:

h (m)	correction (°)	$\bar{s}$ before (mm)	$\bar{s}$ after (mm)	reduction
0.394	−1.622	−28.77	−1.16	96.0 %
0.399	−1.974	−34.53	−1.20	96.5 %
0.404	**−1.461**	**−26.96**	**−2.02**	**92.5** %
0.409	−0.943	−19.29	−2.83	85.3 %
0.414	−1.295	−25.11	−2.92	88.4 %

The controller's own position error falls to ≤0.12 mm at every height and the anchor integral to 0.4 % of its clamp; what remains is the 1 – 3 mm frame-convention gap. The required correction is not a constant (it spans 0.94 – 1.97° over 2 cm), so this is a per-height recalibration of the feedforward map rather than a single scalar.

Both corrections were then put through the full push protocol of §V-B (N=10 repetitions × 8 bearings, binary search over 10 – 160 N at 5 N tolerance; Welch's t against the published arm):

Arm	$F_{\min}$ (N)	F_{med} (N)
ACC as published (k_i=4)	82.8(10)	116.8(47)
k_i=40	81.8(18) p=0.14	116.8(70) p=0.98
k_i=160	81.9(27) p=0.32	115.3(36) p=0.43
Feedforward pitch recalibrated	82.5(12) p=0.55	112.4(49) p=0.05

The published arm reproduces the 82.8 N weakest bearing of §V-B exactly, validating the harness. Raising k_i by 40× costs nothing measurable on either statistic and removes 80 % of the offset; the feedforward recalibration removes 92.5 % and leaves $F_{\min}$ unchanged, but shows a 4.5 N (3.8 %) reduction in F_{med} at p=0.051, which at this sample size we report as unresolved rather than as null. Neither correction is promoted into the shipped profile for this paper, because every table reported here was measured on the published configuration and adopting a change would invalidate that evidence base; they are reported as measured, reproducible routes to closing the limitation.

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